SEGMENTASI CITRA

November 2026

TOPIK

- Pendahuluan
- Edge based:
  - Mendeteksi discontinuities
  - Edge Linking and Boundary detection
- Region based:
  - Thresholding
  - Region Growing
  - Region Merging and Splitting
  - Clustering
- Hybrid

PENDAHULUAN

SEGMENTASI ?

Segmentasi merupakan proses mempartisi citra menjadi beberapa daerah atau objek

Segmentasi citra pada umumnya berdasar pada sifat discontinuity atau similarity dari intensitas piksel

tidak ada kebenaran dasar untuk tugas segmentasi (contoh diberikan pada slide berikutnya)

SEGMENTASI CITRA

Segmentasi mempartisi citra R menjadi sub-region $R_1, R_2, \ldots, R_n$ sehingga:

- $R_1 \cup R_2 \cup \cdots R_n = R$
- Setiap R_i adalah himpunan terhubung
- $R_i \cup R_j = 0$ untuk semua i,j dimana $i \neq j$
- $Q(R_i) = TRUE$ untuk setiap i
- $Q\left(R_i \cup R_j\right) = FALSE$ untuk sembarang dua region yang berdekatan

DILEMMA

input

result 1

result 2

Apa yang kita maksud dengan objek "BERBEDA?

Contoh lain:
ketika kita melihat pohon dari jarak dekat, kita menganggap masing-masing sebagai objek yang berbeda; tetapi ketika kita melihat pohon-pohon jauh, mereka bergabung menjadi satu objek yang koheren (kayu)

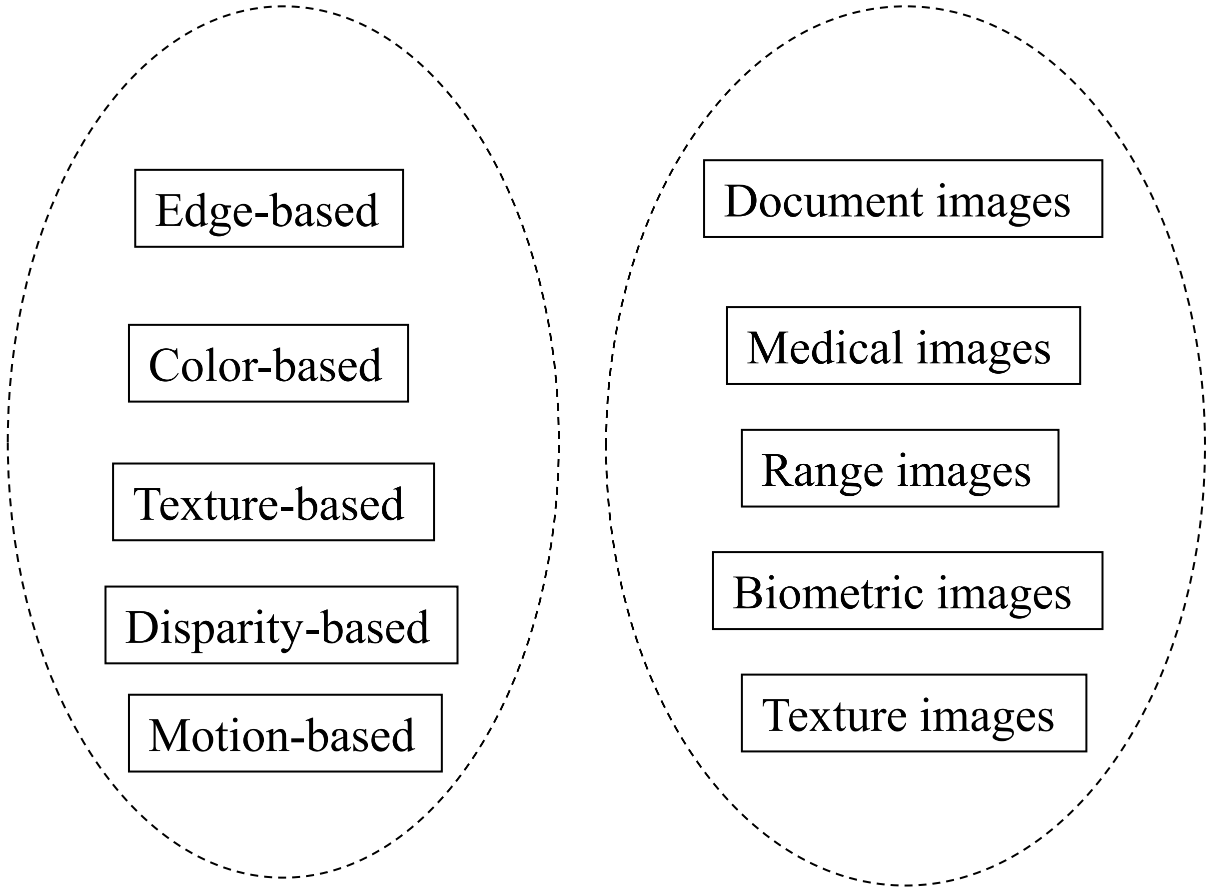
TEKNIK SEGMENTASI

Pendekatan discontinuity:

- mempartisi citra bila terdapat perubahan intensitas secara tiba-tiba (edge based)

Pendekatan similarity:

- mempartisi citra menjadi daerah-daerah yang memiliki kesamaan sifat tertentu (region based)
- contoh: thresholding, region growing, region splitting and merging



MENDETEKSI DISCONTINUITY

Terdiri dari: deteksi titik, deteksi garis, deteksi sisi

Dapat menggunakan mask/kernel, berbeda untuk setiap jenis deteksi

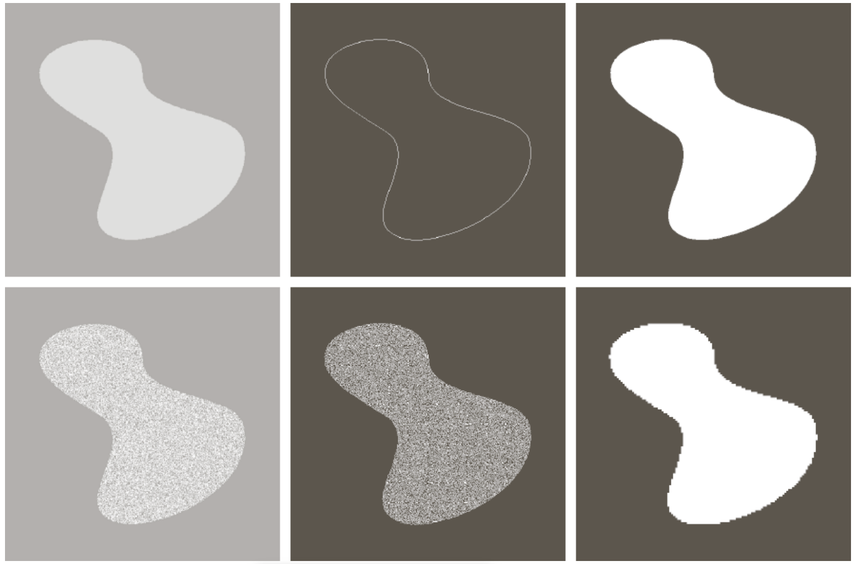
Khusus untuk deteksi sisi, dapat menggunakan operator gradien (Roberts, Sobel, Prewitt), atau Laplacian.

Segmentasi Citra

Citra Asli

Edge-based segmentation

Region-based segmentation



Citra tersegmen R_i aadalah orthogonal dan mempunyai intensitas berbeda

Citra terrsegmen berdasarkn upon standar devisiasi dari intensitas wilayah 4x4

Batas objek diperoleh dari turunan atau Teknik yang mirip

a b c
d e f

FIGURE 10.1 (a) Image containing a region of constant intensity. (b) Image showing the boundary of the inner region, obtained from intensity discontinuities. (c) Result of segmenting the image into two regions. (d) Image containing a textured region. (e) Result of edge computations. Note the large number of small edges that are connected to the original boundary, making it difficult to find a unique boundary using only edge information. (f) Result of segmentation based on region properties.

LATAR BELAKANG

Turunan orde-1

$$\frac{\partial f}{\partial x} = f'(x) = f(x+1) - f(x)$$

Turunan orde-2

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

a b
c

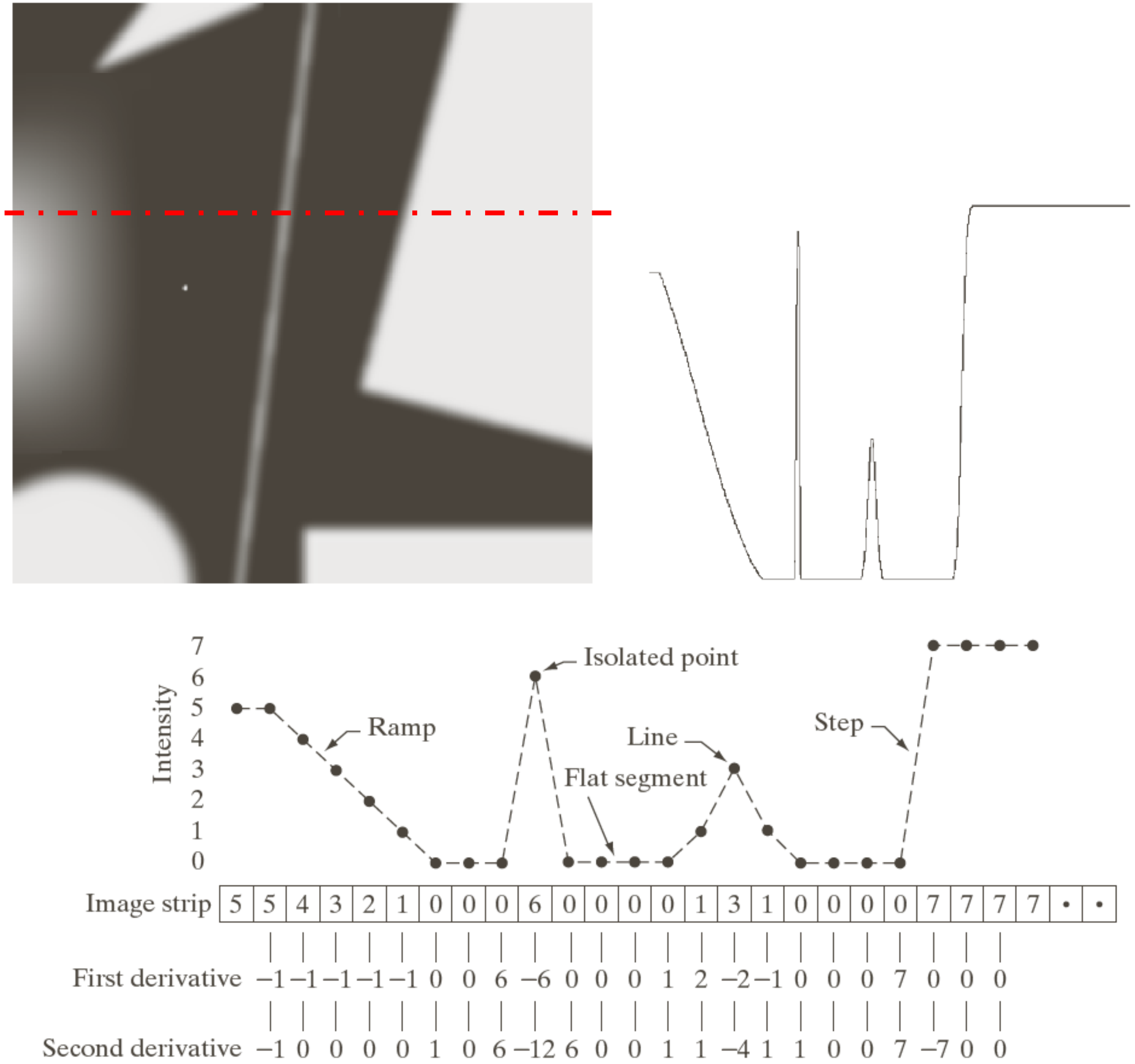


FIGURE 10.2 (a) Image. (b) Horizontal intensity profile through the center of the image, including the isolated noise point. (c) Simplified profile (the points are joined by dashes for clarity). The image strip corresponds to the intensity profile, and the numbers in the boxes are the intensity values of the dots shown in the profile. The derivatives were obtained using Eqs. (10.2-1) and (10.2-2).

KARAKTERISTIK TURUNAN KE-1 DAN KE-2

Derivatif orde pertama umumnya menghasilkan tepi yang lebih tebal pada gambar

Derivatif orde kedua memiliki respons yang lebih kuat terhadap detail halus, seperti garis tipis, titik terisolasi, dan noise

Derivatif orde kedua menghasilkan respons sisi ganda pada transisi ramp dan langkah dalam intensitas

Tanda turunan kedua dapat digunakan untuk menentukan apakah transisi ke tepi adalah dari terang ke gelap atau gelap ke terang

DETEKSI TITIK TERISOLASI

The Laplacian

$$\nabla^2 f(x,y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

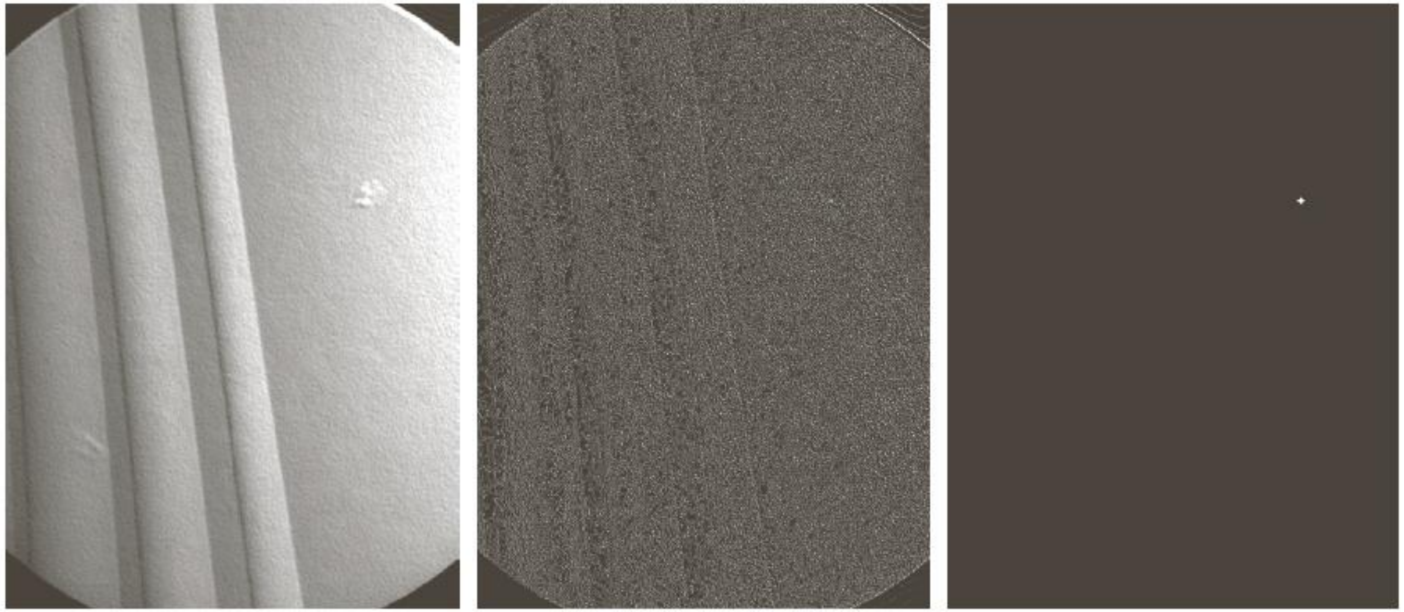
$$= f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1)$$

$$-4f(x,y)$$

$$g(x,y) = \begin{cases} 1 & \text{if } |R(x,y)| \geq T \\ 0 & \text{otherwise} \end{cases}$$

$$R = \sum_{k=1}^{9} w_k z_k$$

1	1	1
1	−8	1
1	1	1



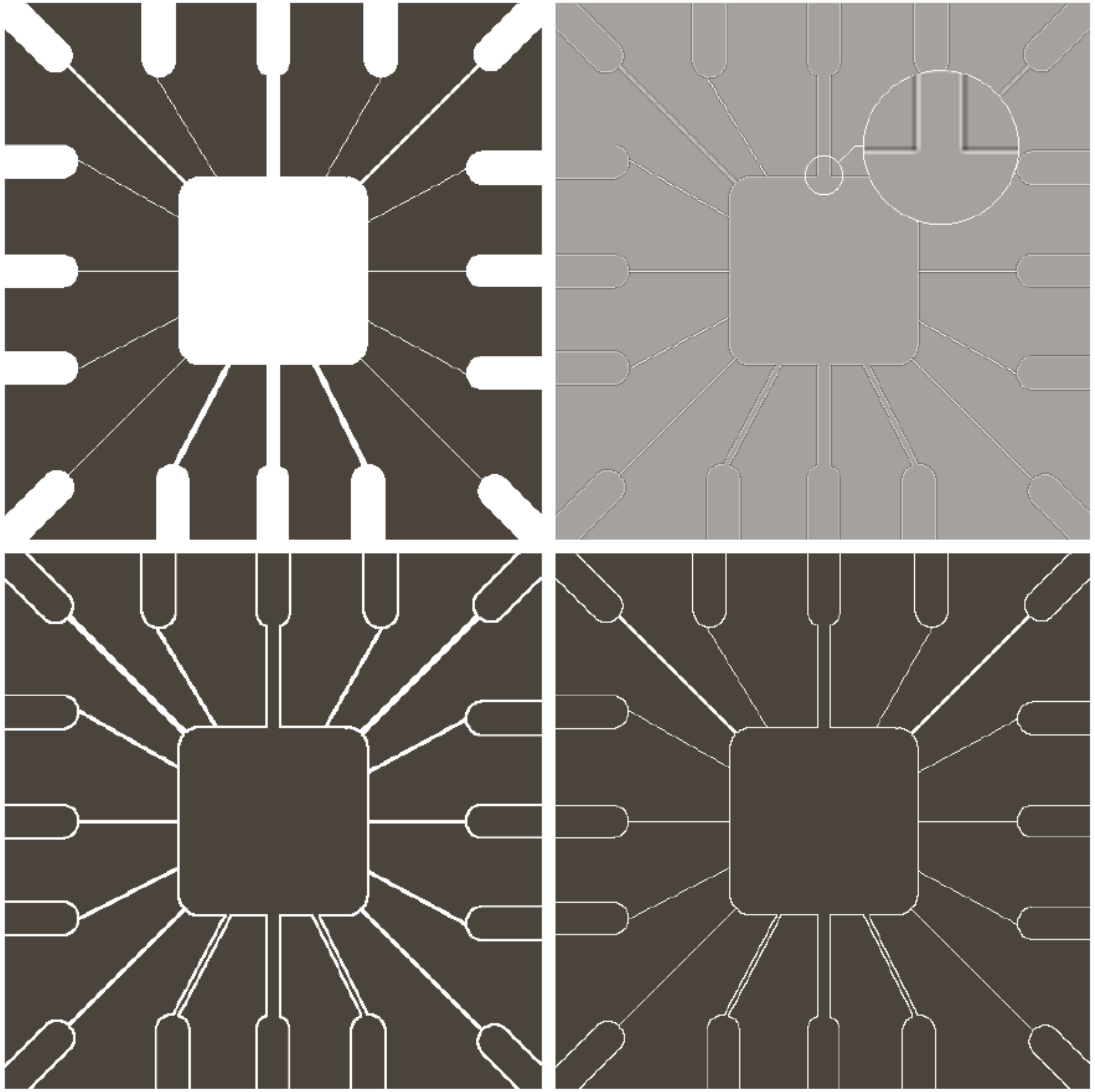
a
b c d

FIGURE 10.4 (a) Point detection (Laplacian) mask. (b) X-ray image of turbine blade with a porosity. The porosity contains a single black pixel. (c) Result of convolving the mask with the image. (d) Result of using Eq. (10.2-8) showing a single point (the point was enlarged to make it easier to see). (Original image courtesy of X-TEK Systems, Ltd.)

DETEKSI GARIS (LINE DETECTION)

Turunan kedua untuk menghasilkan respons yang lebih kuat dan menghasilkan garis yang lebih tipis daripada turunan pertama

Efek garis ganda dari turunan kedua harus ditangani dengan benar



a	b
c	d

FIGURE 10.5
(a) Original image.
(b) Laplacian image; the magnified section shows the positive/negative double-line effect characteristic of the Laplacian.
(c) Absolute value of the Laplacian.
(d) Positive values of the Laplacian.

DETEKSI GARIS PADA ARAH TERTENTU

−1	−1	−1
2	2	2
−1	−1	−1

Horizontal

2	−1	−1
−1	2	−1
−1	−1	2

+45°

−1	2	−1
−1	2	−1
−1	2	−1

Vertical

−1	−1	2
−1	2	−1
2	−1	−1

−45°

FIGURE 10.6 Line detection masks. Angles are with respect to the axis system in Fig. 2.18(b).

Misalkan R_1, R_2, R_3, and R_4 dicatat sebagai respon dari mask pada Fig. 10.6. Sebuah titik pada citra dimana, $|R_k|>|R_j|$, for all $j \neq k$, maka titik dikatakan lebih berasosiasi dengan garis pada arah mask-k

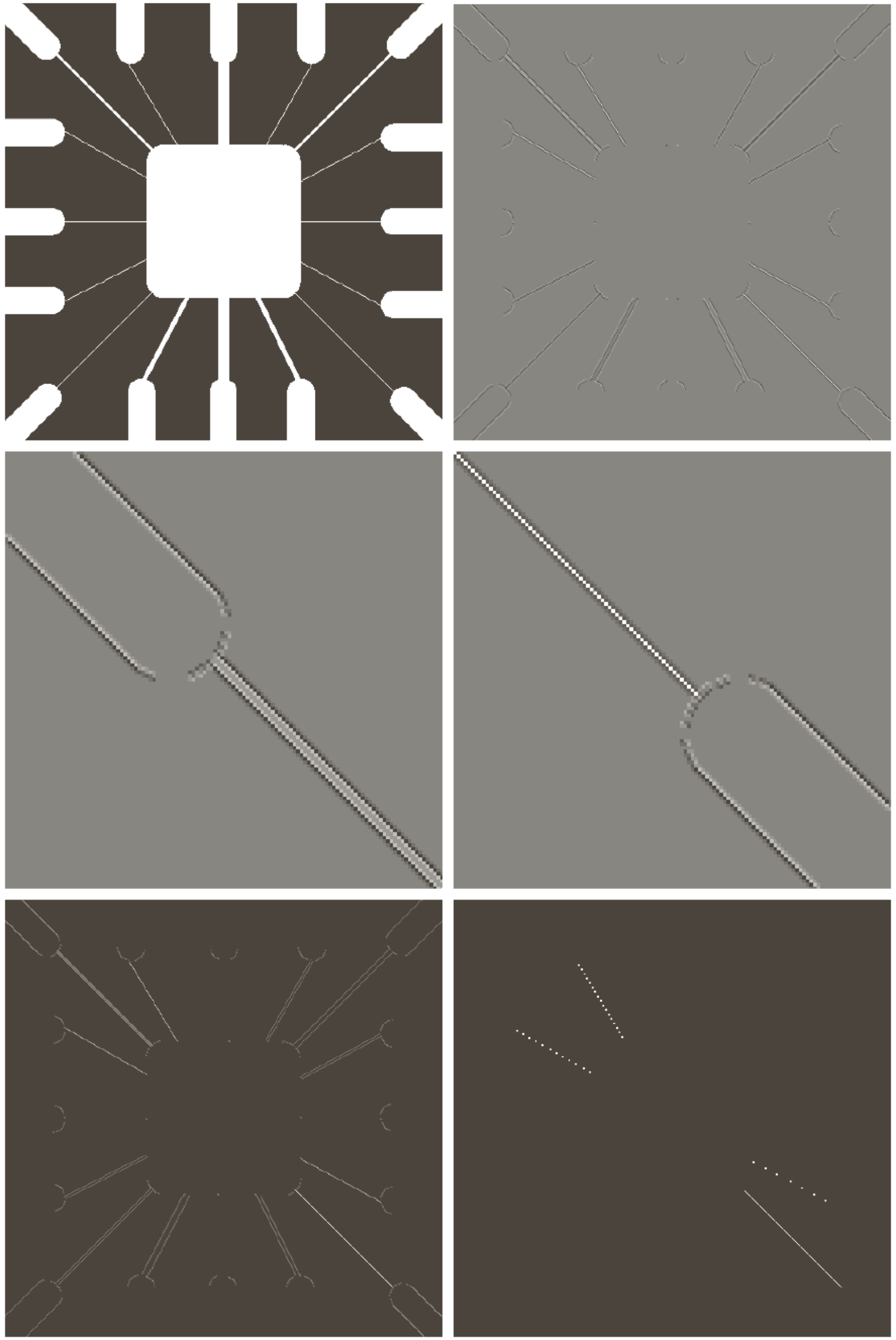
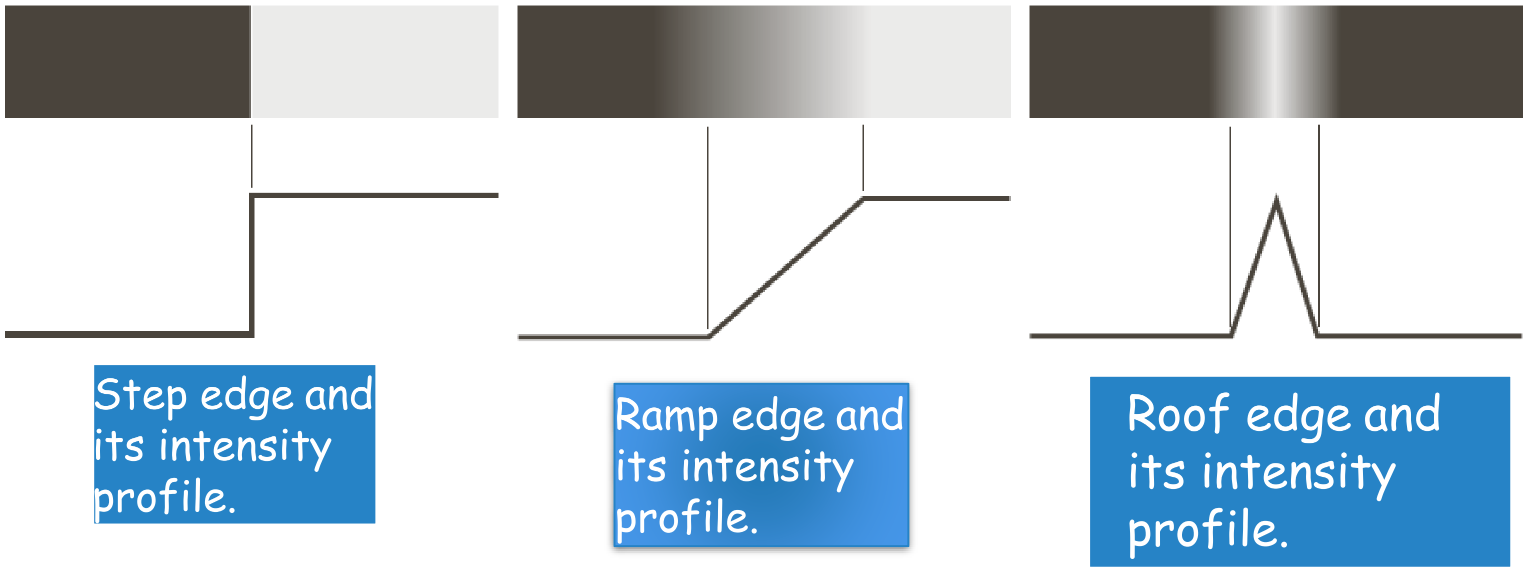


FIGURE 10.7 (a) Image of a wire-bond template. (b) Result of processing with the $+45°$ line detector mask in Fig. 10.6. (c) Zoomed view of the top left region of (b). (d) Zoomed view of the bottom right region of (b). (e) The image in (b) with all negative values set to zero. (f) All points (in white) whose values satisfied the condition $g \geq T$, where g is the image in (e). (The points in (f) were enlarged to make them easier to see.)

DETEKSI TEPI (EDGE DETECTION)

Tepi adalah piksel di mana fungsi kecerahan berubah secara tiba-tiba

Model tepi



a b c

FIGURE 10.8 From left to right, models (ideal representations) of a step, a ramp, and a roof edge, and their corresponding intensity profiles.

PENDEKATAN EDGE-BASED

Kekurangannya: belum tentu menghasilkan edge yang kontinue, mengakibatkan terjadinya kebocoran wilayah (wilayah-wilayah yang tidak tertutup)

Prosedur:

- Melakukan proses deteksi sisi dengan operator gradient. Masukannya citra gray level dan keluarannya citra edge (biner)

Citra intensitas nyata menunjukkan beberapa jenis tepi

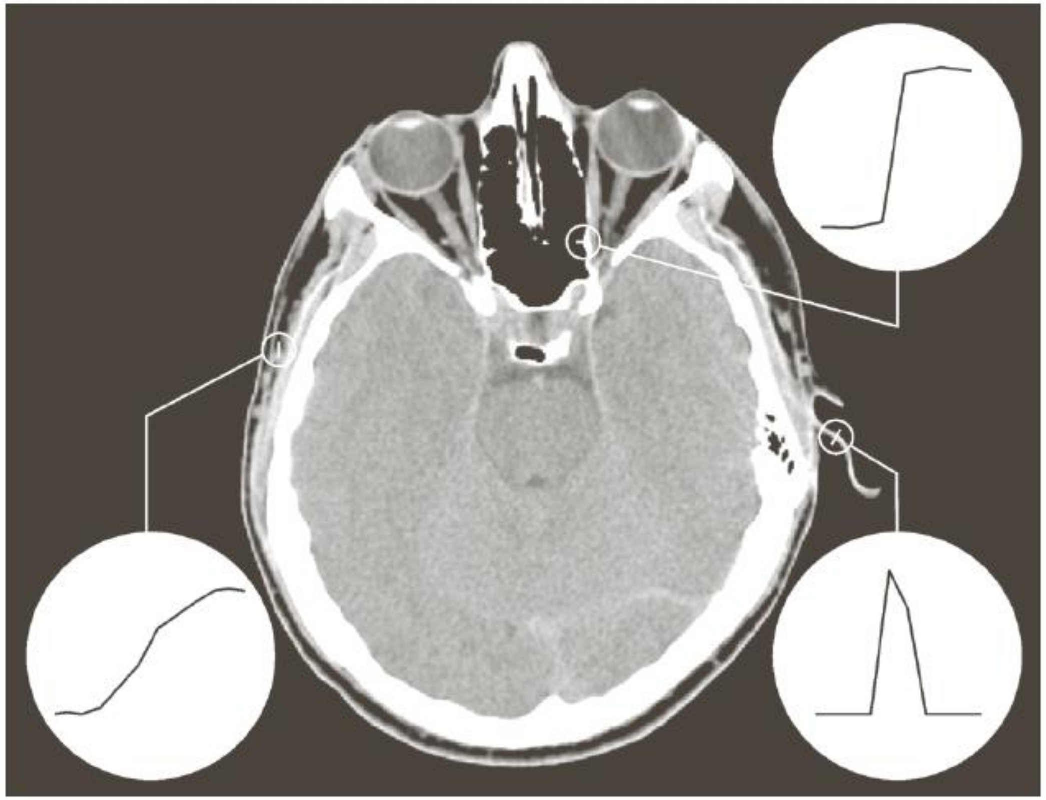
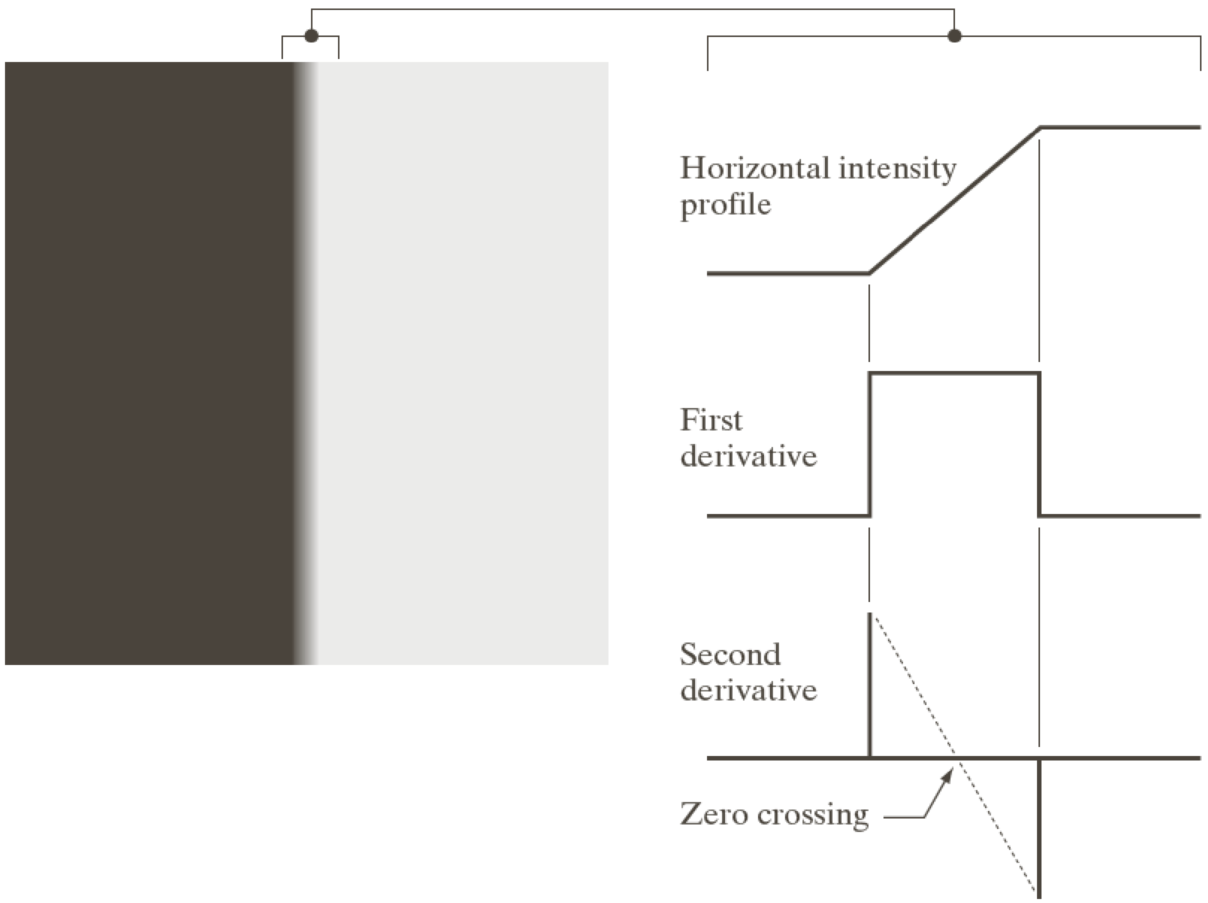


FIGURE 10.9 A 1508 × 1970 image showing (zoomed) actual ramp (bottom, left), step (top, right), and roof edge profiles. The profiles are from dark to light, in the areas indicated by the short line segments shown in the small circles. The ramp and "step" profiles span 9 pixels and 2 pixels, respectively. The base of the roof edge is 3 pixels. (Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)

Pengamatan:

- Besaran turunan pertama dapat digunakan untuk mendeteksi keberadaan sisi pada suatu titik
- Tanda turunan kedua menunjukkan sisi mana dari tepi piksel berada
- Zero crossings dari turunan kedua dapat digunakan untuk menemukan pusat tepi tebal



a b

FIGURE 10.10
(a) Two regions of constant intensity separated by an ideal vertical ramp edge.
(b) Detail near the edge, showing a horizontal intensity profile, together with its first and second derivatives.

Ramp edge corrupted by steadily increasing amounts of additive Gaussian noise

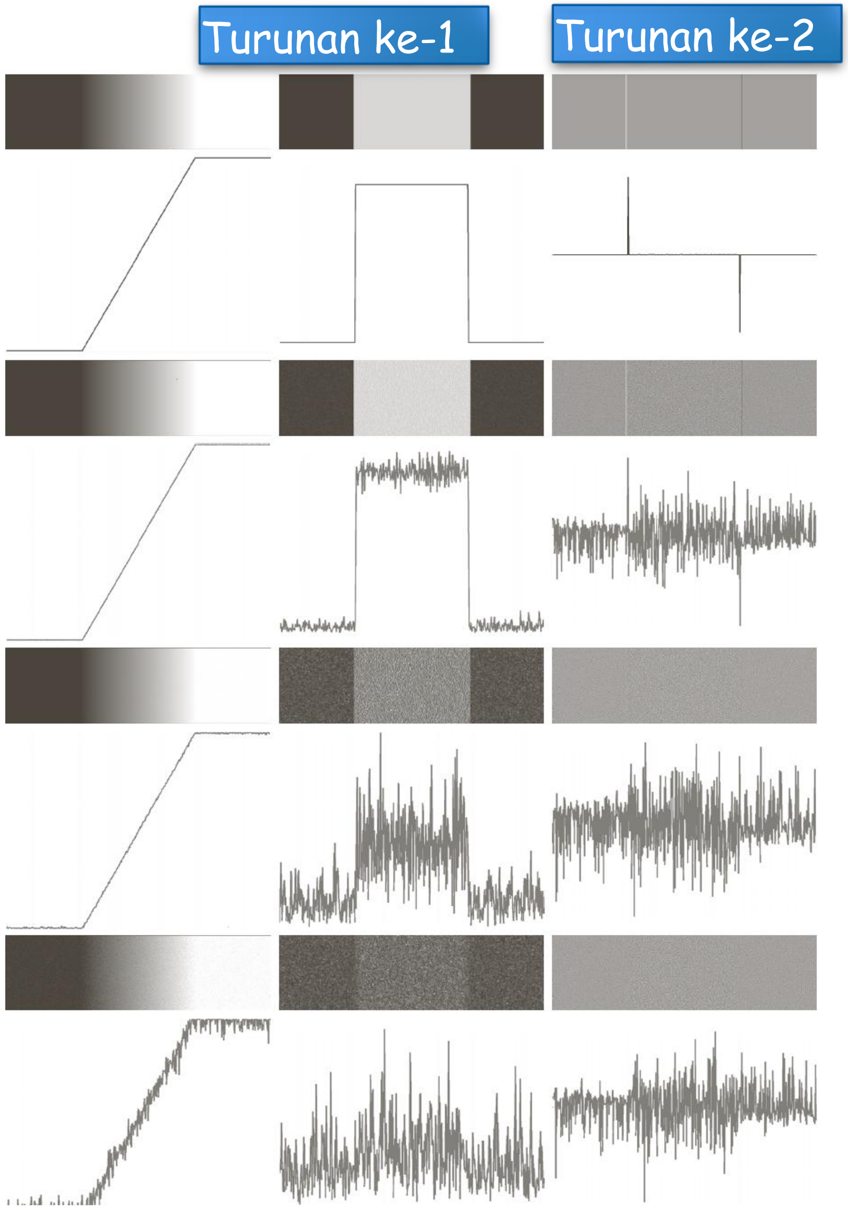


FIGURE 10.11 First column: Images and intensity profiles of a ramp edge corrupted by random Gaussian noise of zero mean and standard deviations of 0.0, 0.1, 1.0, and 10.0 intensity levels, respectively. Second column: First-derivative images and intensity profiles. Third column: Second-derivative images and intensity profiles.

<u>Fundamental steps in edge detection:</u>

1. Smoothing — for noise reduction
2. Detection — of all potential edge candidates
3. Localization — select the real edge points

BASIC EDGE DETECTION BY USING FIRST-ORDER DERIVATIVE

$$\nabla f \equiv grad(f) = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix}$$

The magnitude of ∇f

$$M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^{\ 2} + g_y^{\ 2}}$$

The direction of ∇f

$$\alpha(x, y) = \tan^{-1}\left[\frac{g_x}{g_y}\right]$$

The direction of the edge

$$\phi = \alpha - 90^\circ$$

BASIC EDGE DETECTION BY USING FIRST-ORDER DERIVATIVE

$$\text{Edge normal: } \nabla f \equiv grad(f) = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix}$$

$$\text{Edge unit normal: } \qquad \nabla f \,/\, \mathrm{mag}(\nabla f)$$

In practice,sometimes the magnitude is approximated by

$$\mathrm{mag}(\nabla f) = \left|\frac{\partial f}{\partial x}\right| + \left|\frac{\partial f}{\partial y}\right| \text{ or } \mathrm{mag}(\nabla f) = \max\left(\left|\frac{\partial f}{\partial x}\right|, \left|\frac{\partial f}{\partial y}\right| \right)$$

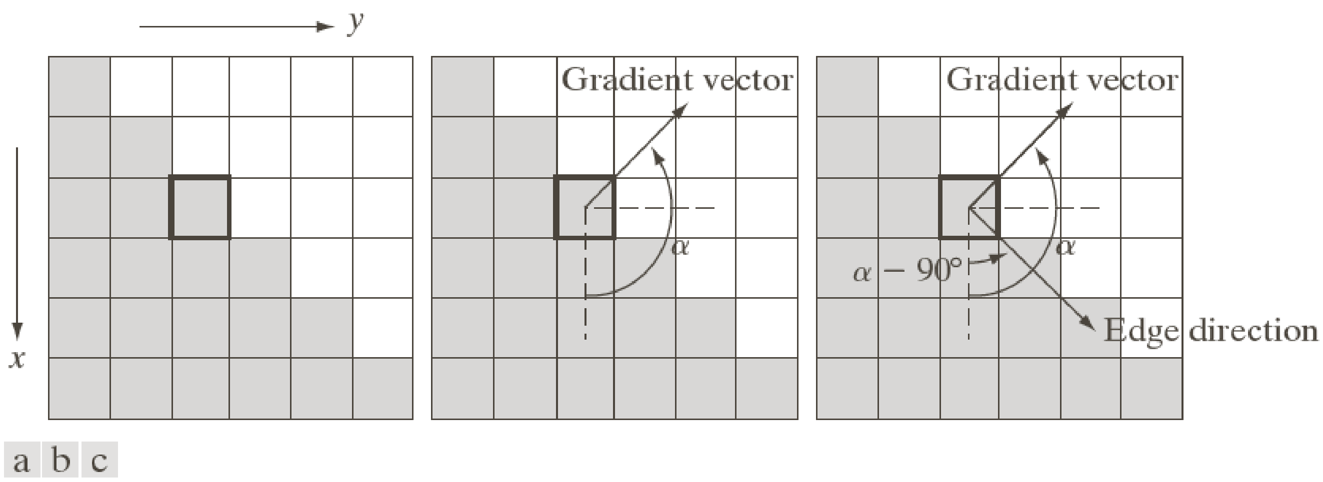


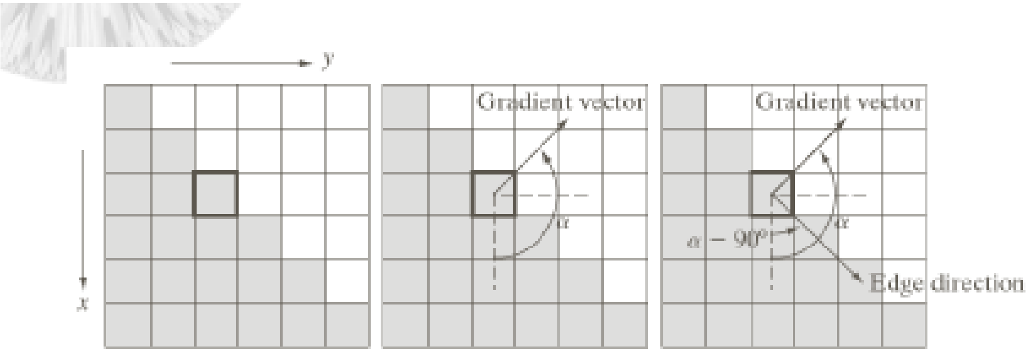
FIGURE 10.12 Using the gradient to determine edge strength and direction at a point. Note that the edge is perpendicular to the direction of the gradient vector at the point where the gradient is computed. Each square in the figure represents one pixel.

$$\nabla f = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} g_x \\ g_y \end{bmatrix}$$

Strength and direction of an edge can be determined using the gradient

Strength (magnitude) $M(x,y) = mag(\nabla f) = \sqrt{g_x^2 + g_y^2}$

Direction $\alpha(x,y) = Tan^{-1}\left(\dfrac{g_x}{g_y}\right)$



$$\nabla f = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Gray=0, white=1

The gradient vector is perpendicular to the actual edge.

−1
1

−1	1

a b

FIGURE 10.13 One-dimensional masks used to implement Eqs. (10.2-12) and (10.2-13).

The Prewitt and Sobel are the most commonly used gradient masks

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

−1	0
0	1

0	−1
1	0

Roberts

−1	−1	−1
0	0	0
1	1	1

−1	0	1
−1	0	1
−1	0	1

Prewitt

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1

Sobel

	a
b	c
d	e
f	g

FIGURE 10.14 A 3 × 3 region of an image (the z's are intensity values) and various masks used to compute the gradient at the point labeled z_5.

The Sobel has better noise characteristics than the Prewitt

Modified Prewitt and Sobel masks for detecting diagonal edges

Prewitt

0	1	1
−1	0	1
−1	−1	0

−1	−1	0
−1	0	1
0	1	1

Sobel

0	1	2
−1	0	1
−2	−1	0

−2	−1	0
−1	0	1
0	1	2

-45°

+45°

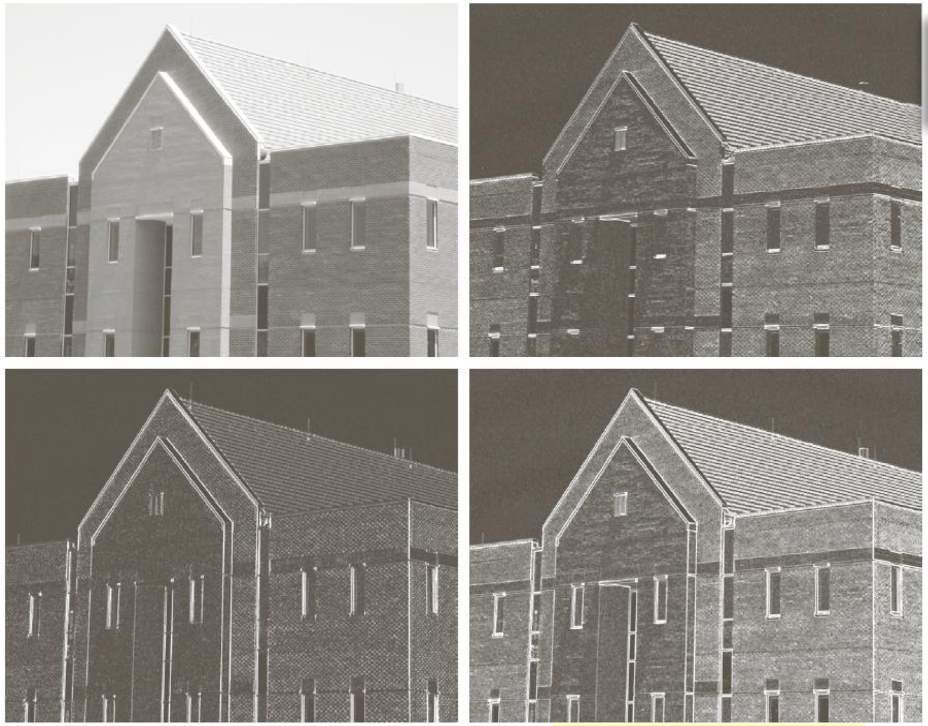
a b
c d

FIGURE 10.15 Prewitt and Sobel masks for detecting diagonal edges.

Image Segmentation

1200x1600 original image

$|G_x|$ using 3x3 Sobel



$|G_y|$ using 3x3 Sobel

$|G_x|+|G_y|$ gradient approximation

a b
c d

FIGURE 10.16 (a) Original image of size 834×1114 pixels, with intensity values scaled to the range [0, 1]. (b) $|g_x|$, the component of the gradient in the x-direction, obtained using the Sobel mask in Fig. 10.14(f) to filter the image. (c) $|g_y|$, obtained using the mask in Fig. 10.14(g). (d) The gradient image, $|g_x| + |g_y|$.

Image Segmentation

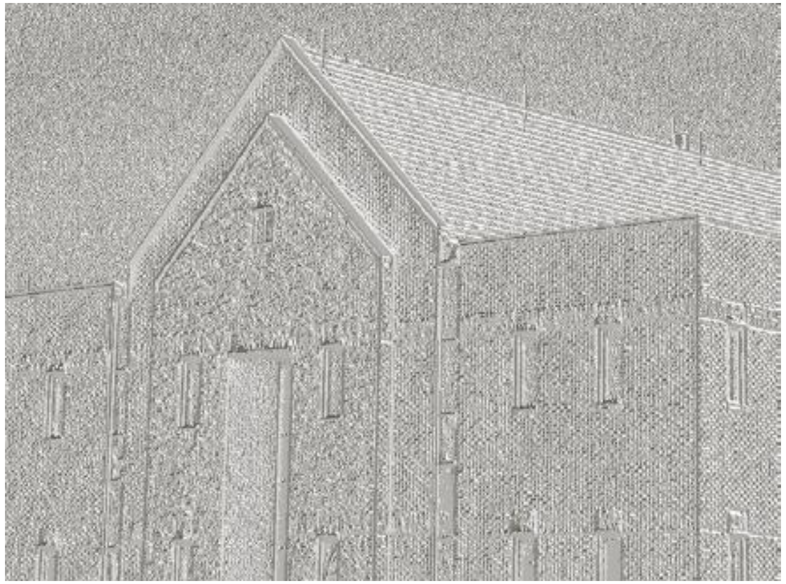
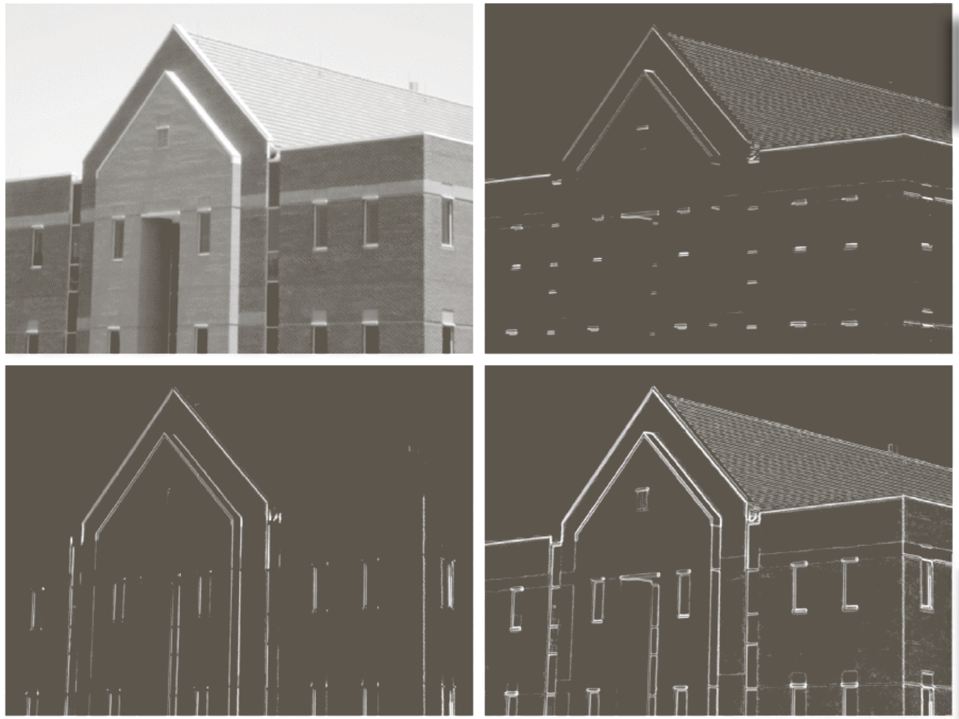


FIGURE 10.17 Gradient angle image computed using Eq. (10.2-11). Areas of constant intensity in this image indicate that the direction of the gradient vector is the same at all the pixel locations in those regions.

Gradient angle plotted as an intensity. Areas of constant intensity have the same gradient vector (perpendicular to the edge). Black means the gradient vector is at 0° so the actual edge is vertical (at -90°)

1200x1600 original image smoothed with a 5x5 averaging filter



$|G_x|$ using 3x3 Sobel

$|G_y|$ using 3x3 Sobel

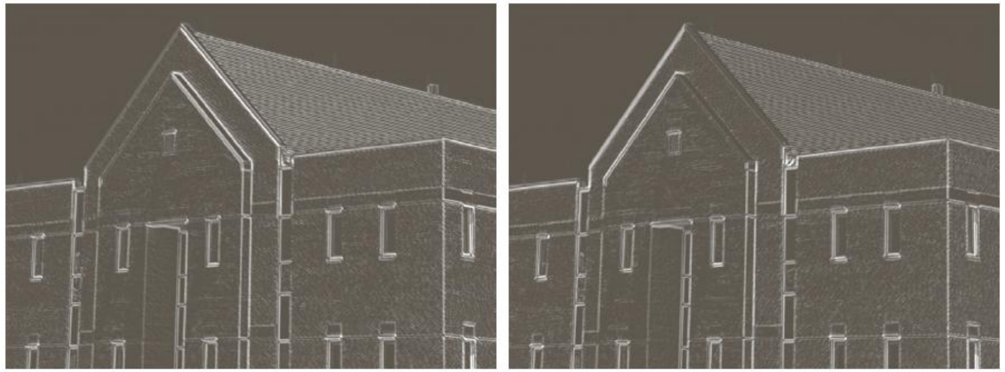
$|G_x|+|G_y|$ gradient approximation

a b
c d

FIGURE 10.18 Same sequence as in Fig. 10.16, but with the original image smoothed using a 5×5 averaging filter prior to edge detection.

Averaging causes the edges to be weaker but cleaner. Note loss of brick edges

Image Segmentation



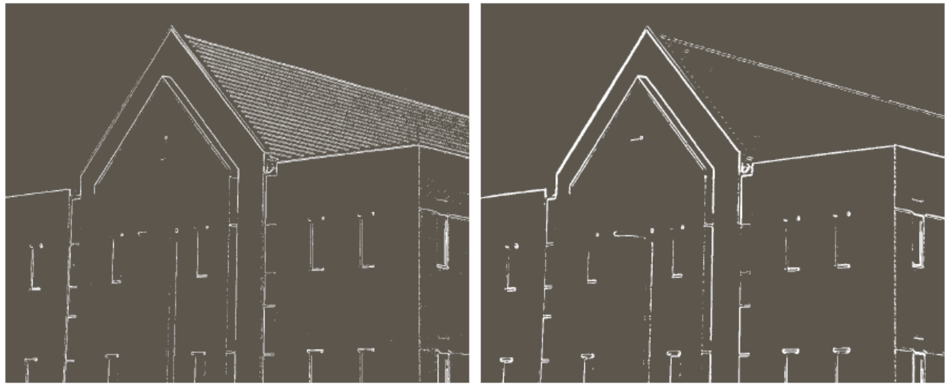
-45° Sobel

+45° Sobel

a b

FIGURE 10.19 Diagonal edge detection. (a) Result of using the mask in Fig. 10.15(c). (b) Result of using the mask in Fig. 10.15(d). The input image in both cases was Fig. 10.18(a).

Note that both can detect a horizontal or vertical edge, but with a weaker response than a horizontal or vertical operator



Thresholded gradient image w/o smoothing

Thresholded gradient image w/smoothing

Threshold = 1/3 max_value

a b

FIGURE 10.20 (a) Thresholded version of the image in Fig. 10.16(d), with the threshold selected as 33% of the highest value in the image; this threshold was just high enough to eliminate most of the brick edges in the gradient image. (b) Thresholded version of the image in Fig. 10.18(d), obtained using a threshold equal to 33% of the highest value in that image.

ADVANCED TECHNIQUES FOR EDGE DETECTION

The Marr-Hildreth edge detector

$$G(x,y) = e^{-\frac{x^2+y^2}{2\sigma^2}}, \quad \sigma : \text{space constant.}$$

Laplacian of Gaussian (LoG)

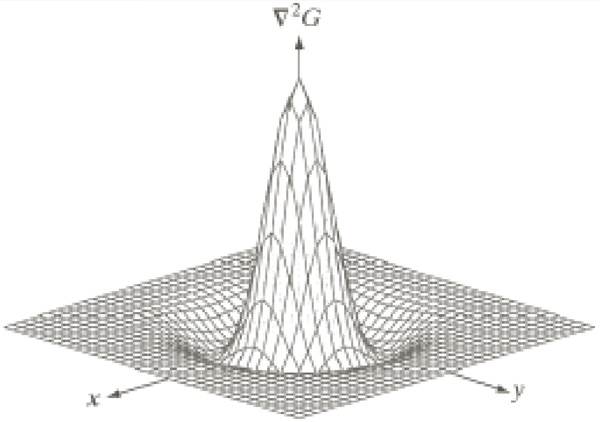
$$\nabla^2 G(x,y) = \frac{\partial^2 G(x,y)}{\partial x^2} + \frac{\partial^2 G(x,y)}{\partial y^2}$$

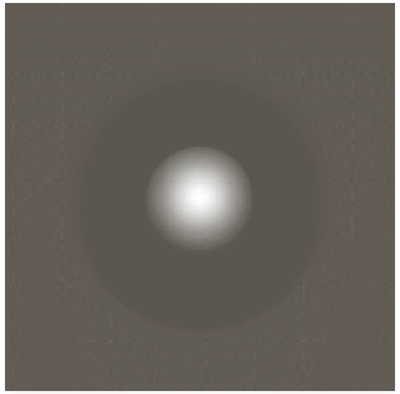
$$= \frac{\partial}{\partial x}\left[\frac{-x}{\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}\right] + \frac{\partial}{\partial y}\left[\frac{-y}{\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}\right]$$

$$= \left[\frac{x^2}{\sigma^4} - \frac{1}{\sigma^2}\right] e^{-\frac{x^2+y^2}{2\sigma^2}} + \left[\frac{y^2}{\sigma^4} - \frac{1}{\sigma^2}\right] e^{-\frac{x^2+y^2}{2\sigma^2}}$$

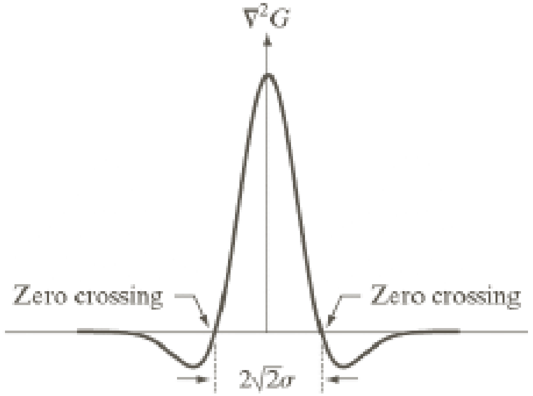
$$= \left[\frac{x^2+y^2-\sigma^2}{\sigma^4}\right] e^{-\frac{x^2+y^2}{2\sigma^2}}$$

The LoG is sometimes called the Mexican hat operator





The Laplacian is NEVER used directly because of its strong noise sensitivity



0	0	−1	0	0
0	−1	−2	−1	0
−1	−2	16	−2	−1
0	−1	−2	−1	0
0	0	−1	0	0

a b
c d

FIGURE 10.21 (a) Three-dimensional plot of the *negative* of the LoG. (b) Negative of the LoG displayed as an image. (c) Cross section of (a) showing zero crossings. (d) 5 × 5 mask approximation to the shape in (a). The negative of this mask would be used in practice.

Combining the Laplacian with a Gaussian gives the LoG

$$\nabla^2 h(r) = -\left[\frac{r^2 - \sigma^2}{\sigma^4}\right] e^{-\frac{r^2}{2\sigma^2}}$$

THE CANNY EDGE DETECTOR: ALGORITHM (1)

Let $f(x, y)$ denote the input image and $G(x, y)$ denote the Gaussian function:

$$G(x, y) = e^{-\frac{x^2+y^2}{2\sigma^2}}$$

We form a smoothed image, $f_s(x, y)$ by convolving G and f:

$$f_s(x, y) = G(x, y) \star f(x, y)$$

THE CANNY EDGE DETECTOR: ALGORITHM(2)

Compute the gradient magnitude and direction (angle):

$$M(x,y) = \sqrt{g_x^{\,2} + g_y^{\,2}}$$

and

$$\alpha(x,y) = \arctan(g_y / g_x)$$

where $g_x = \partial f_s / \partial x$ and $g_y = \partial f_s / \partial y$

Note: any of the filter mask pairs in Fig.10.14 can be used to obtain g_x and g_y

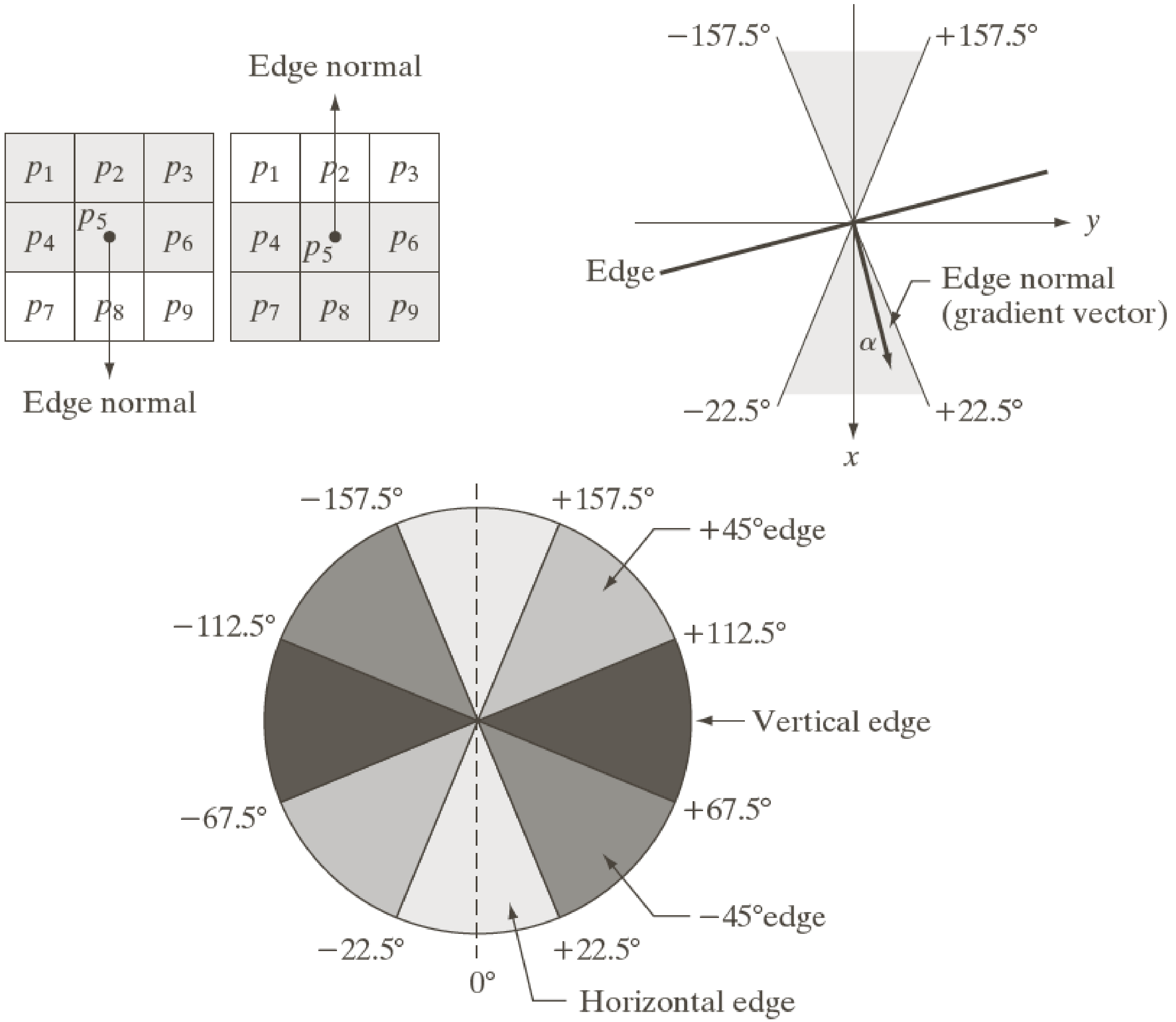
THE CANNY EDGE DETECTOR: ALGORITHM(3)

The gradient $M(x, y)$ typically contains wide ridge around local maxima. Next step is to thin those ridges.

Nonmaxima suppression:

Let d_1, d_2, d_3, and d_4 denote the four basic edge directions for a 3×3 region: horizontal, -45°, vertical, +45°, respectively.

1. Find the direction d_k that is closest to $\alpha(x, y)$.
2. If the value of $M(x, y)$ is less than at least one of its two neighbors along d_k, let $g_N(x, y) = 0$ (suppression); otherwise, let $g_N(x, y) = M(x, y)$



a b
c

FIGURE 10.24
(a) Two possible orientations of a horizontal edge (in gray) in a 3 × 3 neighborhood. (b) Range of values (in gray) of α, the direction angle of the *edge normal*, for a horizontal edge. (c) The angle ranges of the edge normals for the four types of edge directions in a 3 × 3 neighborhood. Each edge direction has two ranges, shown in corresponding shades of gray.

THE CANNY EDGE DETECTOR: ALGORITHM(4)

The final operation is to threshold $g_N(x, y)$ to reduce false edge points.

Hysteresis thresholding:

$$g_{NH}(x, y) = g_N(x, y) \geq T_H$$
$$g_{NL}(x, y) = g_N(x, y) \geq T_L$$

and

$$g_{NL}(x, y) = g_{NL}(x, y) - g_{NH}(x, y)$$

THE CANNY EDGE DETECTOR: ALGORITHM(5)

Depending on the value of T_H, the edges in $g_{NH}(x, y)$ typically have gaps. Longer edges are formed using the following procedure:

(a). Locate the next unvisited edge pixel, p, in $g_{NH}(x, y)$.

(b). Mark as valid edge pixel all the weak pixels in $g_{NL}(x, y)$ that are connected to p using 8-connectivity.

(c). If all nonzero pixel in $g_{NH}(x, y)$ have been visited go to step (d), esle return to (a).

(d). Set to zero all pixels in $g_{NL}(x, y)$ that were not marked as valid edge pixels.

THE CANNY EDGE DETECTION: SUMMARY

Haluskan gambar input dengan filter Gaussian

Hitung besar gradien dan gambar sudut

Terapkan penekanan nonmaxima ke gambar magnitudo gradien

Gunakan ambang batas ganda dan analisis konektivitas untuk mendeteksi dan menghubungkan tepi

a b
c d

FIGURE 10.25 (a) Original image of size 834 × 1114 pixels, with intensity values scaled to the range [0, 1]. (b) Thresholded gradient of smoothed image. (c) Image obtained using the Marr-Hildreth algorithm. (d) Image obtained using the Canny algorithm. Note the significant improvement of the Canny image compared to the other two.

$T_L = 0.04; T_H = 0.10; \sigma = 4$ and a mask of size 25×25

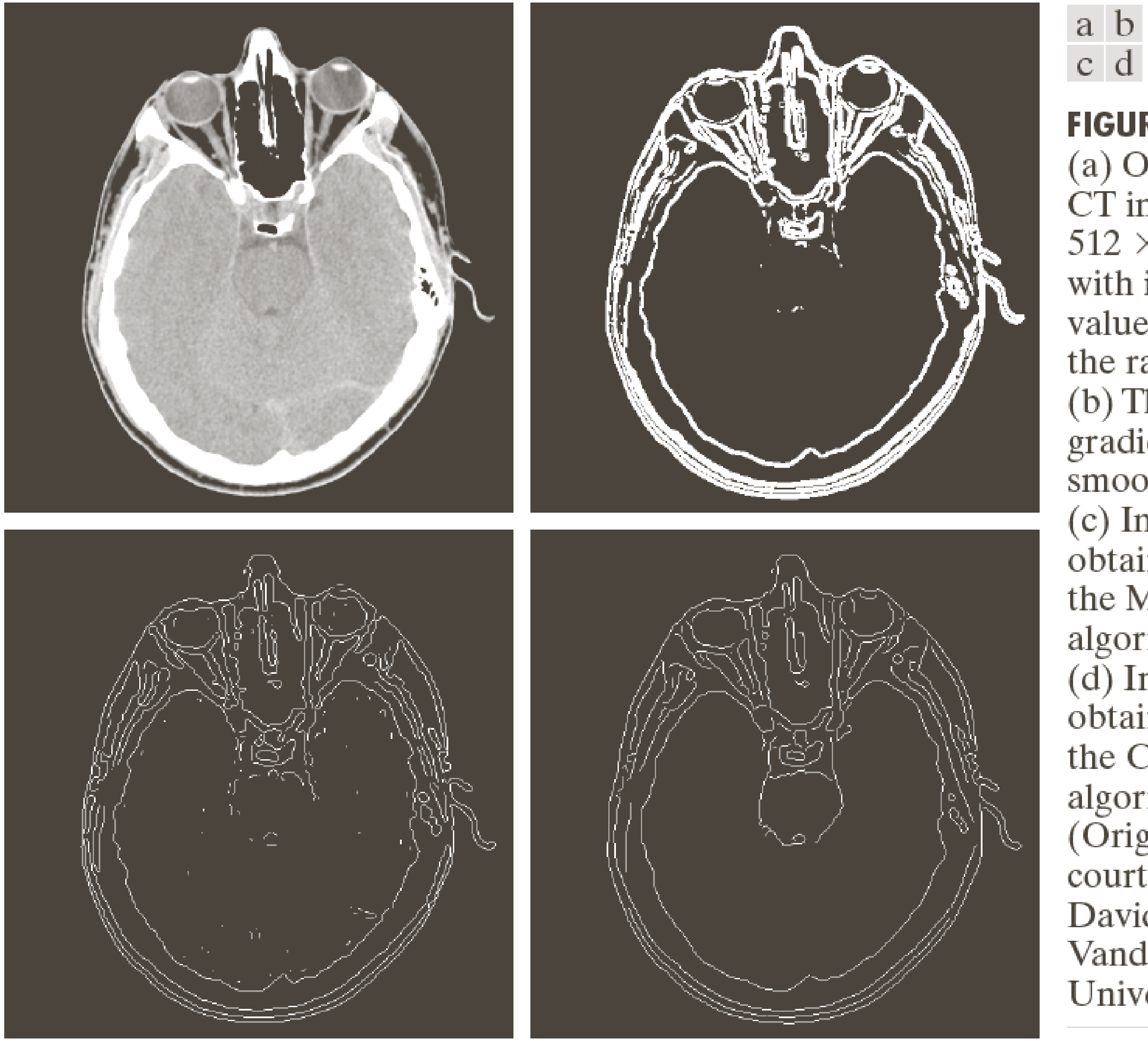
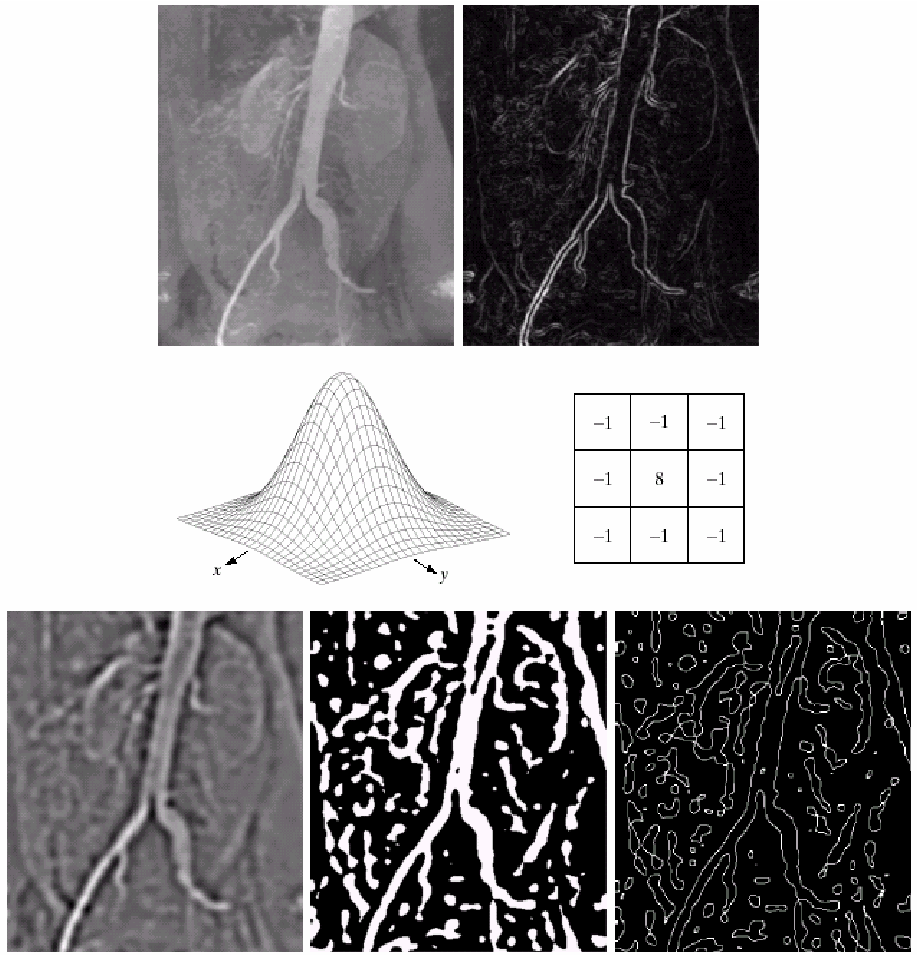


FIGURE 10.26 (a) Original head CT image of size 512 × 512 pixels, with intensity values scaled to the range [0, 1]. (b) Thresholded gradient of smoothed image. (c) Image obtained using the Marr-Hildreth algorithm. (d) Image obtained using the Canny algorithm. (Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)

DETEKSI TEPI



a b
c d
e f g

FIGURE 10.15 (a) Original image. (b) Sobel gradient (shown for comparison). (c) Spatial Gaussian smoothing function. (d) Laplacian mask. (e) LoG. (f) Thresholded LoG. (g) Zero crossings. (Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

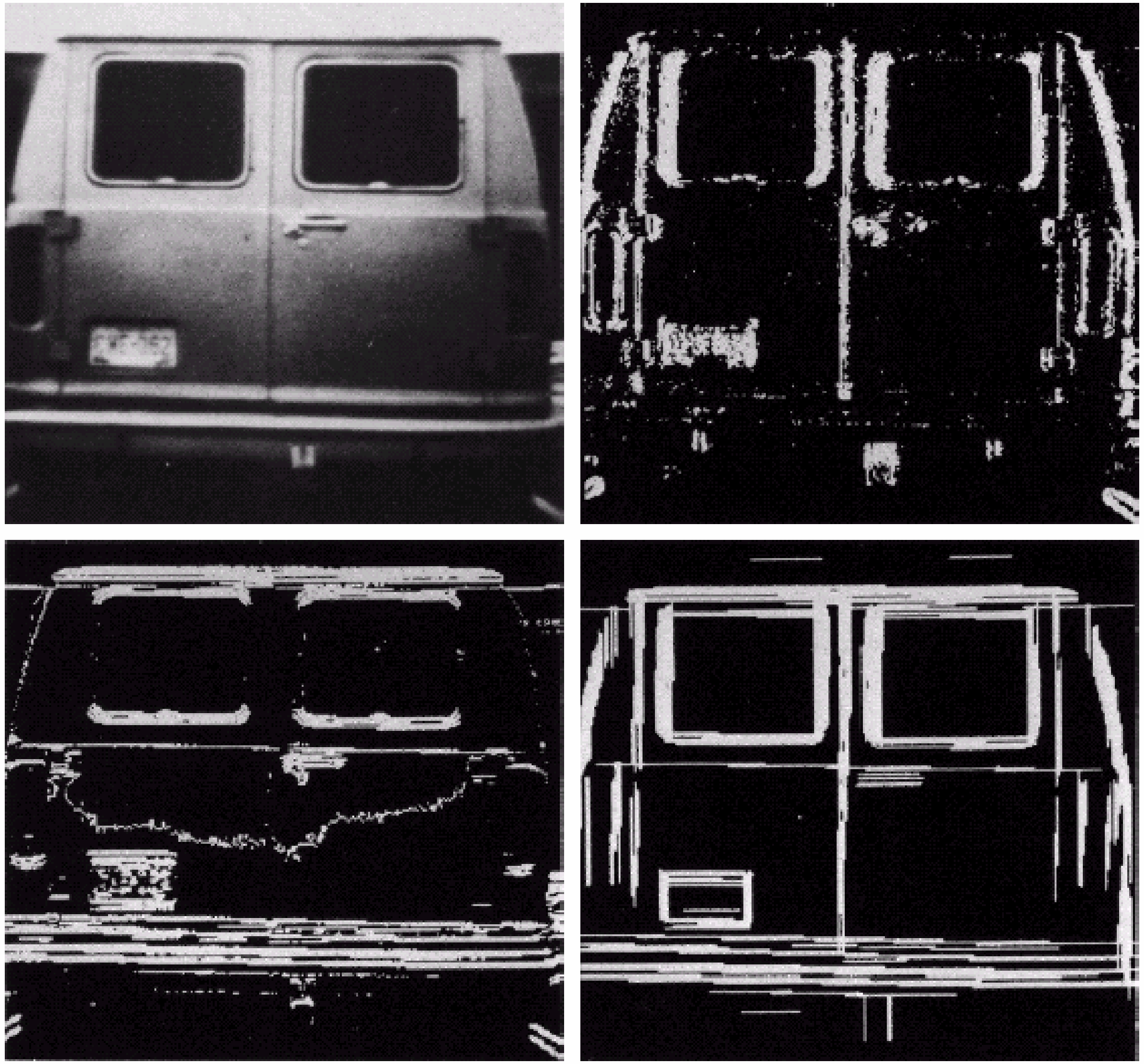
EDGE LINKING & BOUNDARY DET.

- Hasil dari deteksi sisi seringkali tidak menghasilkan sisi yang lengkap, karena adanya noise, patahnya sisi karena iluminasi, dan lain-lain.
- Oleh karena itu proses deteksi sisi biasanya dilanjutkan dengan proses edge linking.
- Dari sekian banyak cara, yang akan dibahas:
  - Local Processing
  - Global Processing dengan teknik Graph-Theoretic

LOCAL PROCESSING

- Merupakan cara paling sederhana, dengan menggunakan analisa ketetanggaan 3x3 atau 5x5.
- Sifat utama yang digunakan untuk menentukan kesamaan piksel sisi adalah
  - Nilai gradien: $\nabla f(x,y) = |G_x| + [G_y|$
  - Arah gradien: $\alpha(x,y) = \tan^{-1}(G_x/G_y)$
- Jika piksel pada posisi (x_0,y_0) adalah piksel sisi, maka piksel pada posisi (x,y) dalam jendela ketetanggaan yang sama bisa dikategorikan sisi pula jika
  - $|\nabla f(x,y) - \nabla f(x_0,y_0)| \leq E$
  - $|\alpha (x,y) - \alpha (x_0,y_0)| < A$
  - E dan A adalah nilai ambang non negatif

CONTOH



a b
c d

FIGURE 10.16
(a) Input image.
(b) G_y component of the gradient.
(c) G_x component of the gradient.
(d) Result of edge linking. (Courtesy of Perceptics Corporation.)

REGION BASED:

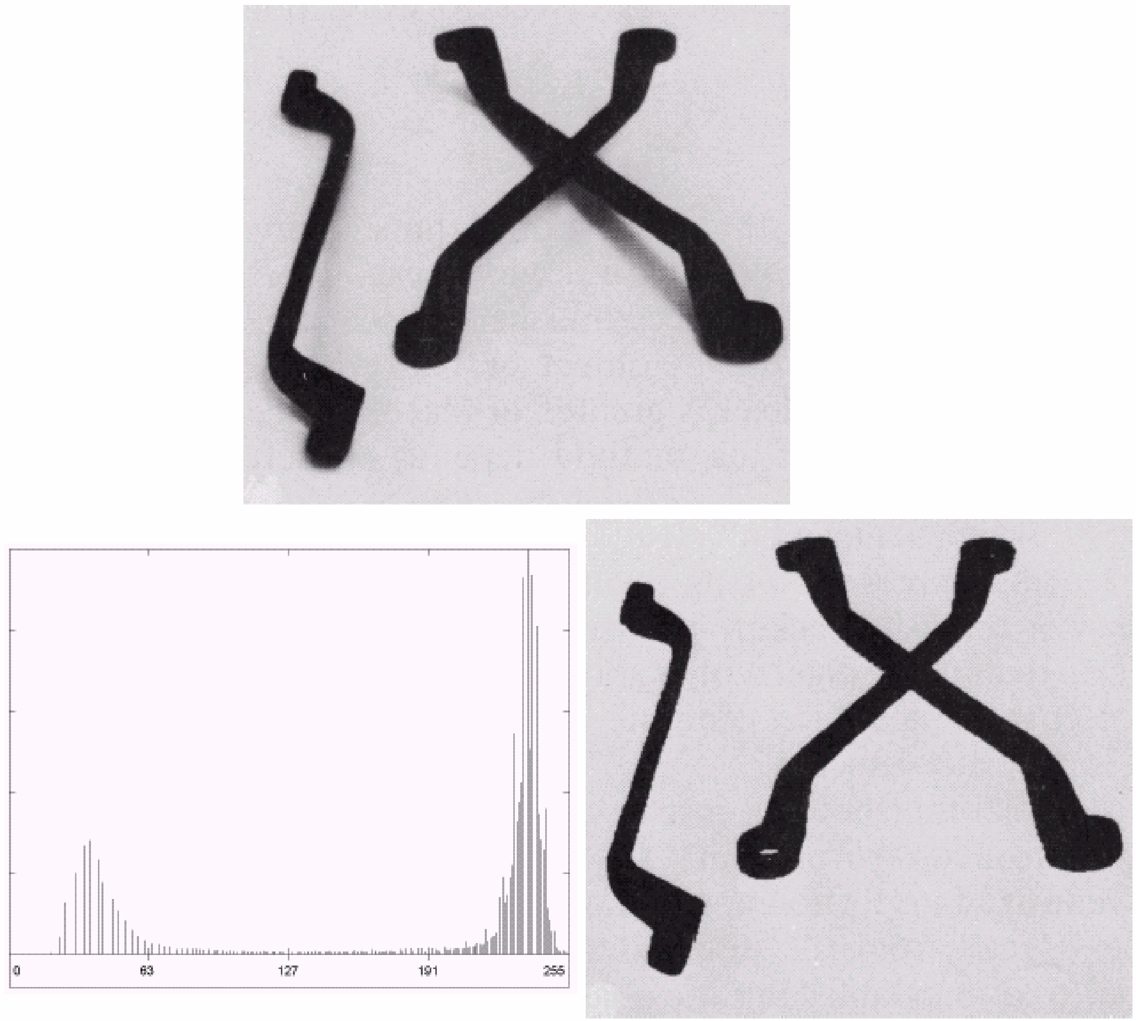
Thresholding
Region Growing
Region Merging and Splitting
Clustering

THRESHOLDING

Sering digunakan untuk segmentasi karena mudah dan intuitif.

Diasumsikan setiap objek cenderung memiliki warna yang homogen dan terletak pada kisaran keabuan tertentu

CONTOH



a
b c

FIGURE 10.28 (a) Original image. (b) Image histogram. (c) Result of global thresholding with T midway between the maximum and minimum gray levels.

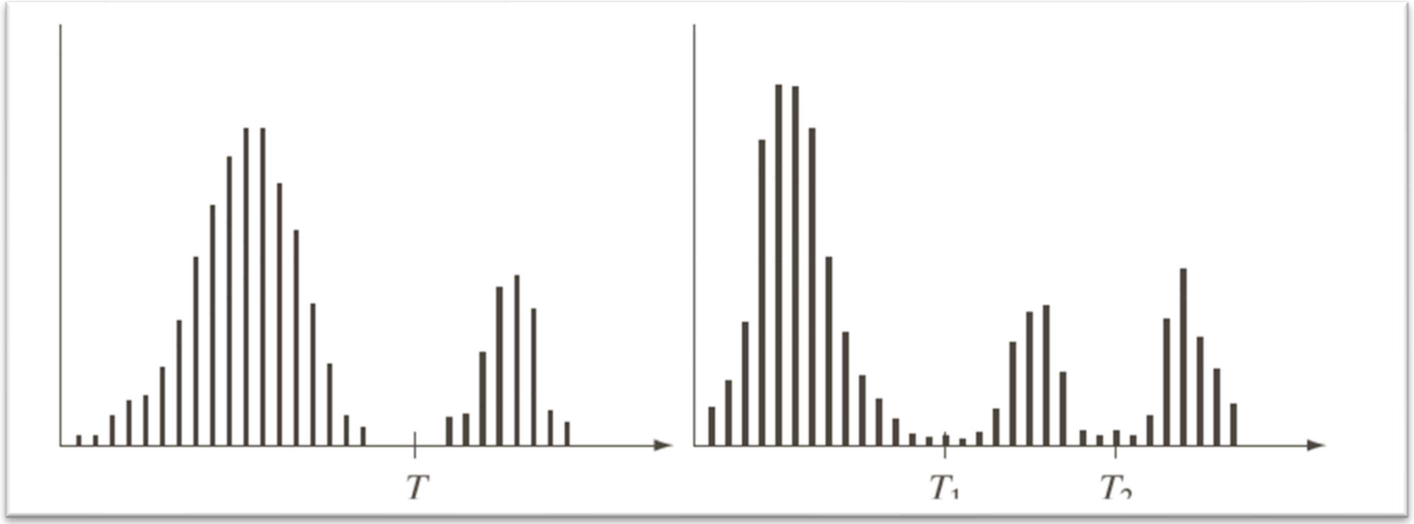
THRESHOLDING

$$g(x,y)=\begin{cases}1 & \text{if } f(x,y)>T \quad \text{(object point)}\\ 0 & \text{if } f(x,y)\le T \quad \text{(background point)}\end{cases}$$

T : global thresholding

Multiple thresholding

$$g(x,y)=\begin{cases}a & \text{if } f(x,y)>T_2\\ b & \text{if } T_1< f(x,y)\le T_2\\ c & \text{if } f(x,y)\le T_1\end{cases}$$



a b

FIGURE 10.35 Intensity histograms that can be partitioned (a) by a single threshold, and (b) by dual thresholds.

THE ROLE OF NOISE IN IMAGE THRESHOLDING

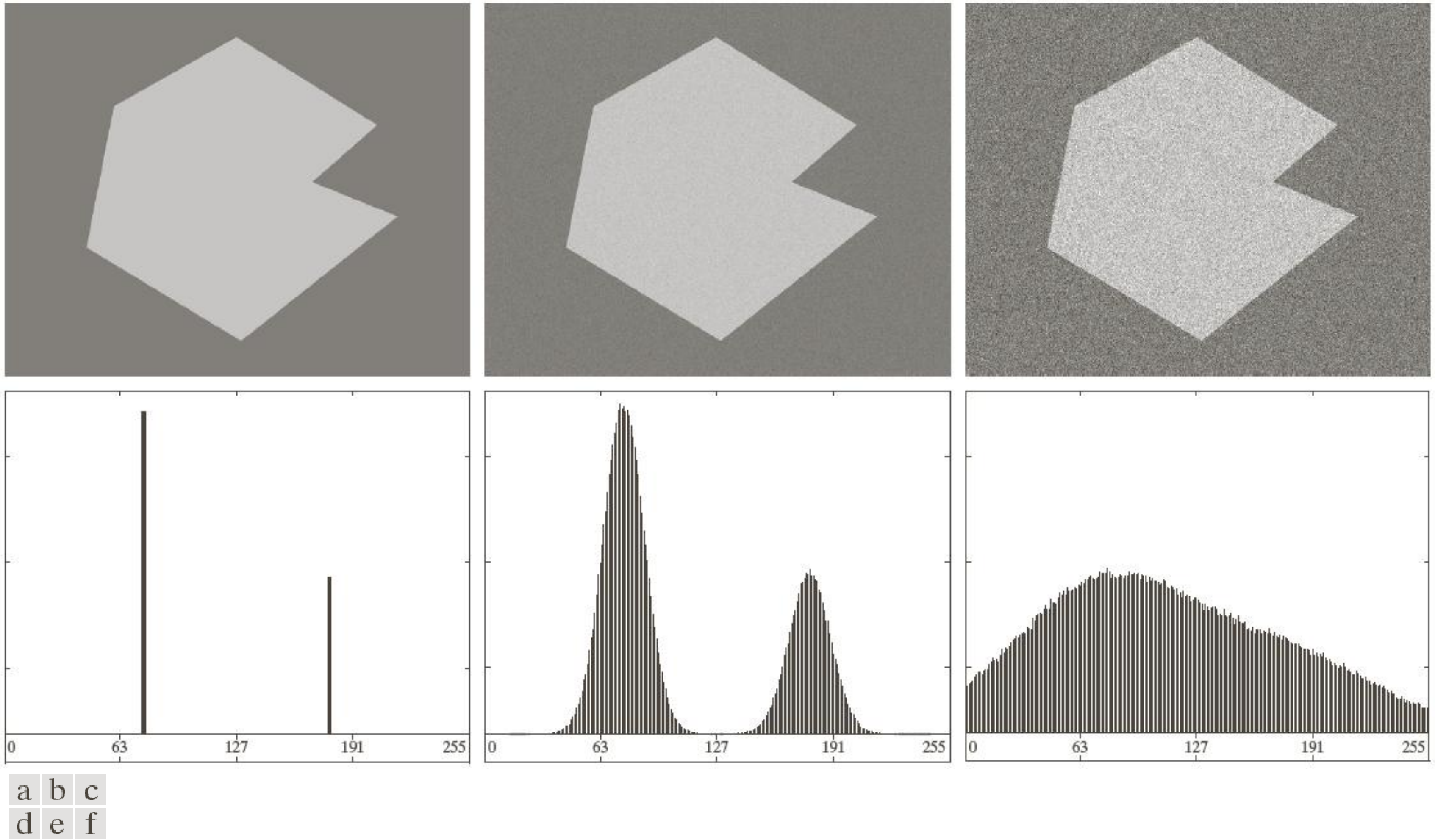


FIGURE 10.36 (a) Noiseless 8-bit image. (b) Image with additive Gaussian noise of mean 0 and standard deviation of 10 intensity levels. (c) Image with additive Gaussian noise of mean 0 and standard deviation of 50 intensity levels. (d)–(f) Corresponding histograms.

THE ROLE OF ILLUMINATION AND REFLECTANCE

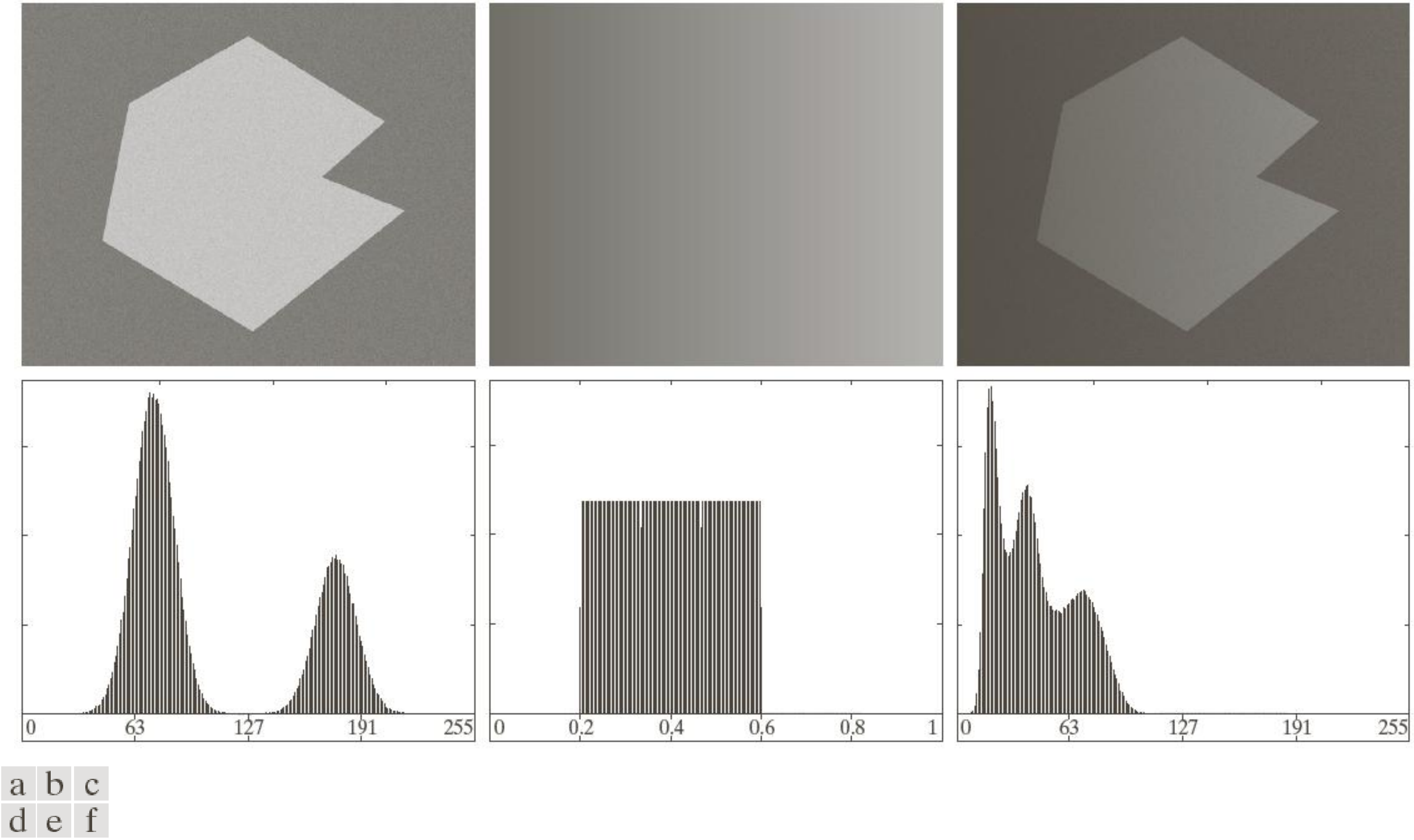


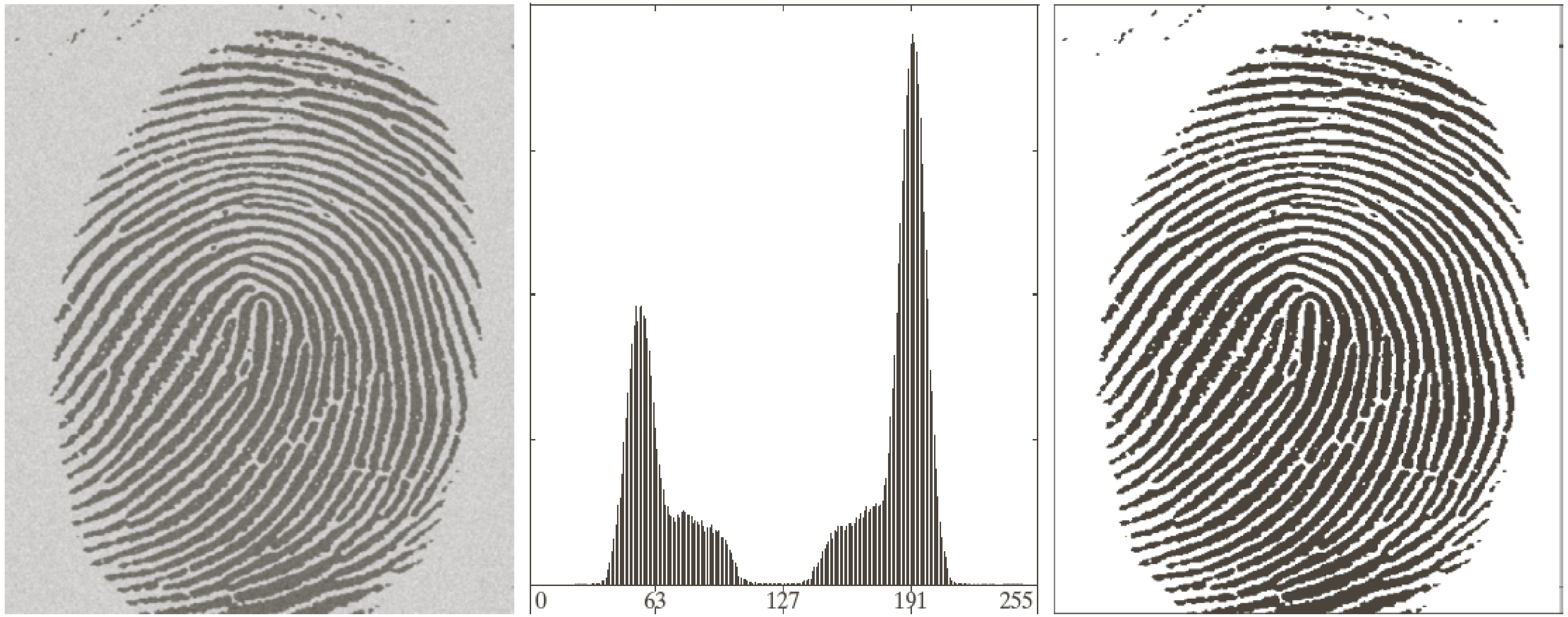
FIGURE 10.37 (a) Noisy image. (b) Intensity ramp in the range [0.2, 0.6]. (c) Product of (a) and (b). (d)–(f) Corresponding histograms.

BASIC GLOBAL THRESHOLDING

1. Select an initial estimate for the global threshold, T.
2. Segment the image using T. It will produce two groups of pixels: G1 consisting of all pixels with intensity values > T and G2 consisting of pixels with values $\leq$ T.
3. Compute the average intensity values m1 and m2 for the pixels in G1 and G2, respectively.
4. Compute a new threshold value.

$$T = \frac{1}{2}\left(m1 + m2\right)$$

5. Repeat Steps 2 through 4 until the difference between values of T in successive iterations is smaller than a predefined parameter ΔT .



a b c

FIGURE 10.38 (a) Noisy fingerprint. (b) Histogram. (c) Segmented result using a global threshold (the border was added for clarity). (Original courtesy of the National Institute of Standards and Technology.)

OPTIMUM GLOBAL THRESHOLDING USING OTSU'S METHOD

Principle: maximizing the between-class variance

Let $\{0,\ 1,\ 2,\ ...,\ L-1\}$ denote the L distinct intensity levels in a digital image of size $M \times N$ pixels, and let n_i denote the number of pixels with intensity i.

$$p_i = n_i / MN \quad \text{and} \quad \sum_{i=0}^{L-1} p_i = 1$$

k is a threshold value, $C_1 \rightarrow [0, k],\ C_2 \rightarrow [k+1, L-1]$

$$P_1(k) = \sum_{i=0}^{k} p_i \quad \text{and} \quad P_2(k) = \sum_{i=k+1}^{L-1} p_i = 1 - P_1(k)$$

OPTIMUM GLOBAL THRESHOLDING USING OTSU'S METHOD

The mean intensity value of the pixels assigned to class C_1 is

$$m_1(k) = \sum_{i=0}^{k} iP(i / C_1) = \frac{1}{P_1(k)} \sum_{i=0}^{k} ip_i$$

The mean intensity value of the pixels assigned to class C_2 is

$$m_2(k) = \sum_{i=k+1}^{L-1} iP(i / C_2) = \frac{1}{P_2(k)} \sum_{i=k+1}^{L-1} ip_i$$

$P_1 m_1 + P_2 m_2 = m_G$ (Global mean value)

OPTIMUM GLOBAL THRESHOLDING USING OTSU'S METHOD

Between-class variance, σ_B^2 is defined as

$$\sigma_B^2 = P_1(m_1 - m_G)^2 + P_2(m_2 - m_G)^2$$

$$= P_1 P_2 (m_1 - m_2)^2$$

$$= \frac{\left[m_G P_1 - m_1 P_1\right]^2}{P_1(1 - P_1)}$$

$$= \frac{\left[m_G P_1 - m\right]^2}{P_1(1 - P_1)}$$

OPTIMUM GLOBAL THRESHOLDING USING OTSU'S METHOD

The optimum threshold is the value, k*, that maximizes

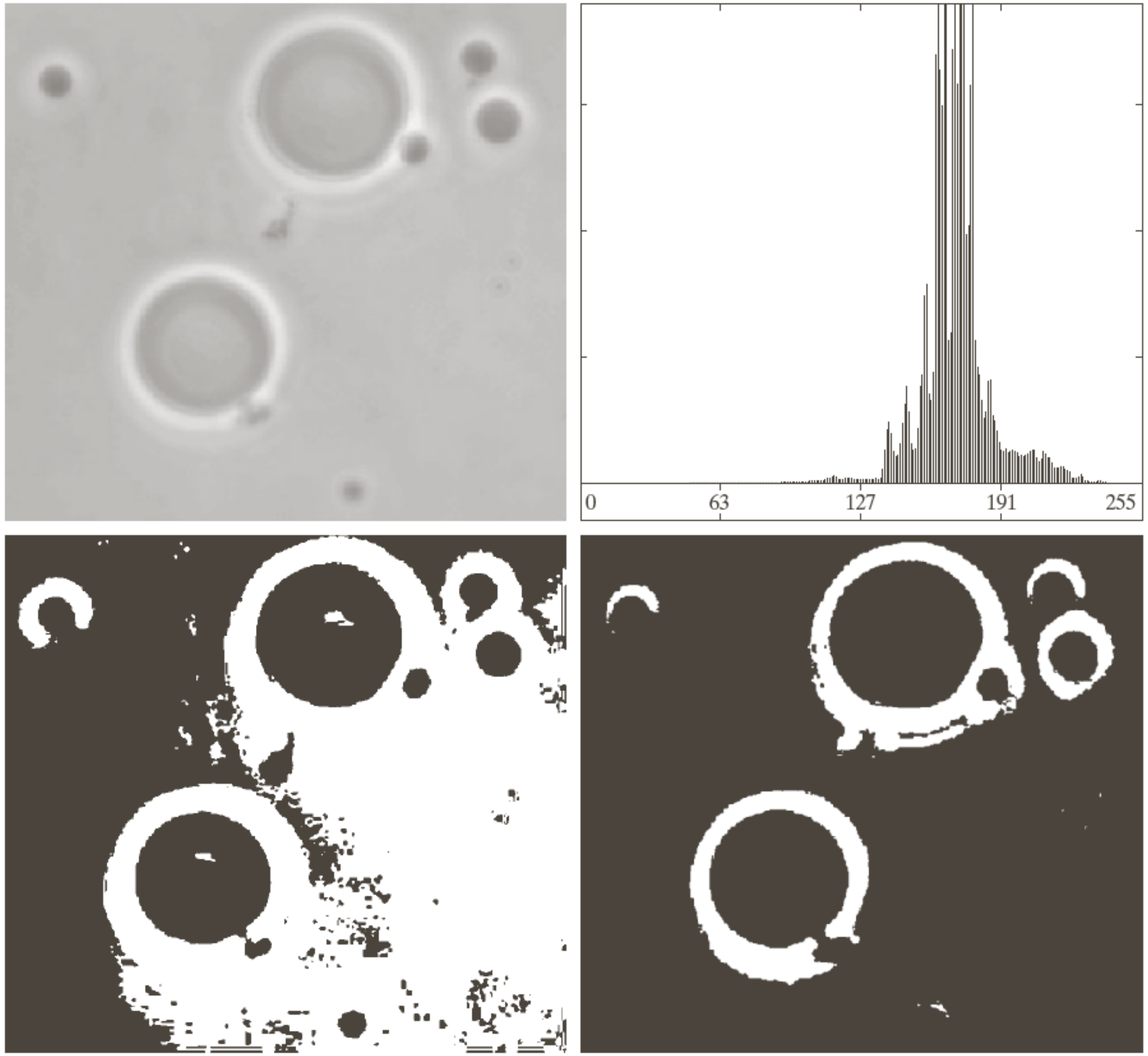
$$\sigma_B^2(k^*), \qquad \sigma_B^2(k^*) = \max_{0 \le k \le L-1} \sigma_B^2(k)$$

$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > k^* \\ 0 & \text{if } f(x, y) \le k^* \end{cases}$$

Separability measure $\eta = \dfrac{\sigma_B^2}{\sigma_G^2}$

OTSU'S ALGORITHM: SUMMARY

1. Compute the normalized histogram of the input image. Denote the components of the histogram by p_i, $i=0, 1, \dots, L-1$.
2. Compute the cumulative sums, $P_1(k)$, for $k = 0, 1, \dots, L-1$.
3. Compute the cumulative means, $m(k)$, for $k = 0, 1, \dots, L-1$.
4. Compute the global intensity mean, m_G.
5. Compute the between-class variance, for $k = 0, 1, \dots, L-1$.
6. Obtain the Otsu's threshold, k^*.
7. Obtain the separability measure.



a b
c d

FIGURE 10.39
(a) Original image.
(b) Histogram (high peaks were clipped to highlight details in the lower values).
(c) Segmentation result using the basic global algorithm from Section 10.3.2.
(d) Result obtained using Otsu's method.
(Original image courtesy of Professor Daniel A. Hammer, the University of Pennsylvania.)

USING IMAGE SMOOTHING TO IMPROVE GLOBAL THRESHOLDING

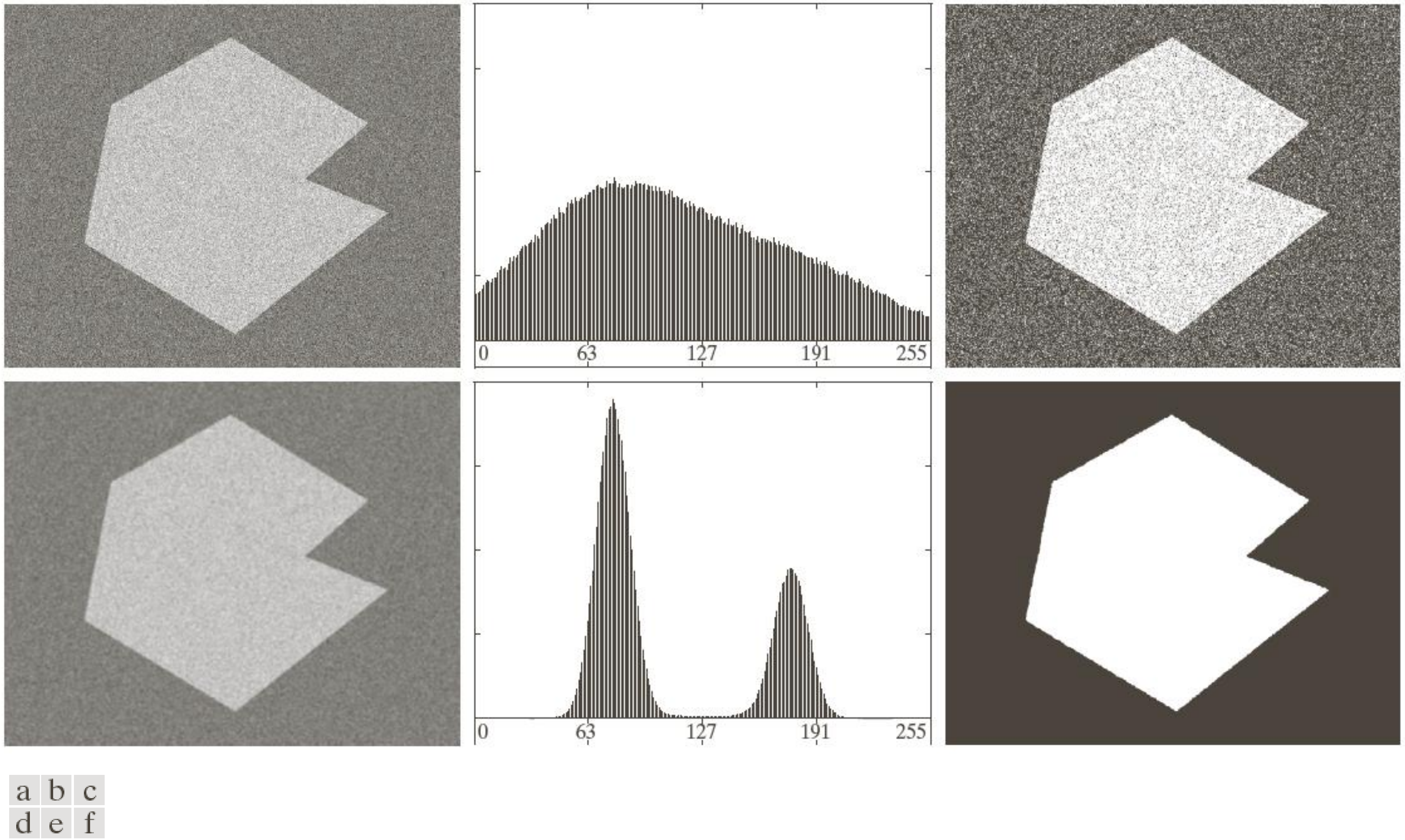


FIGURE 10.40 (a) Noisy image from Fig. 10.36 and (b) its histogram. (c) Result obtained using Otsu's method. (d) Noisy image smoothed using a 5×5 averaging mask and (e) its histogram. (f) Result of thresholding using Otsu's method.

USING EDGES TO IMPROVE GLOBAL THRESHOLDING

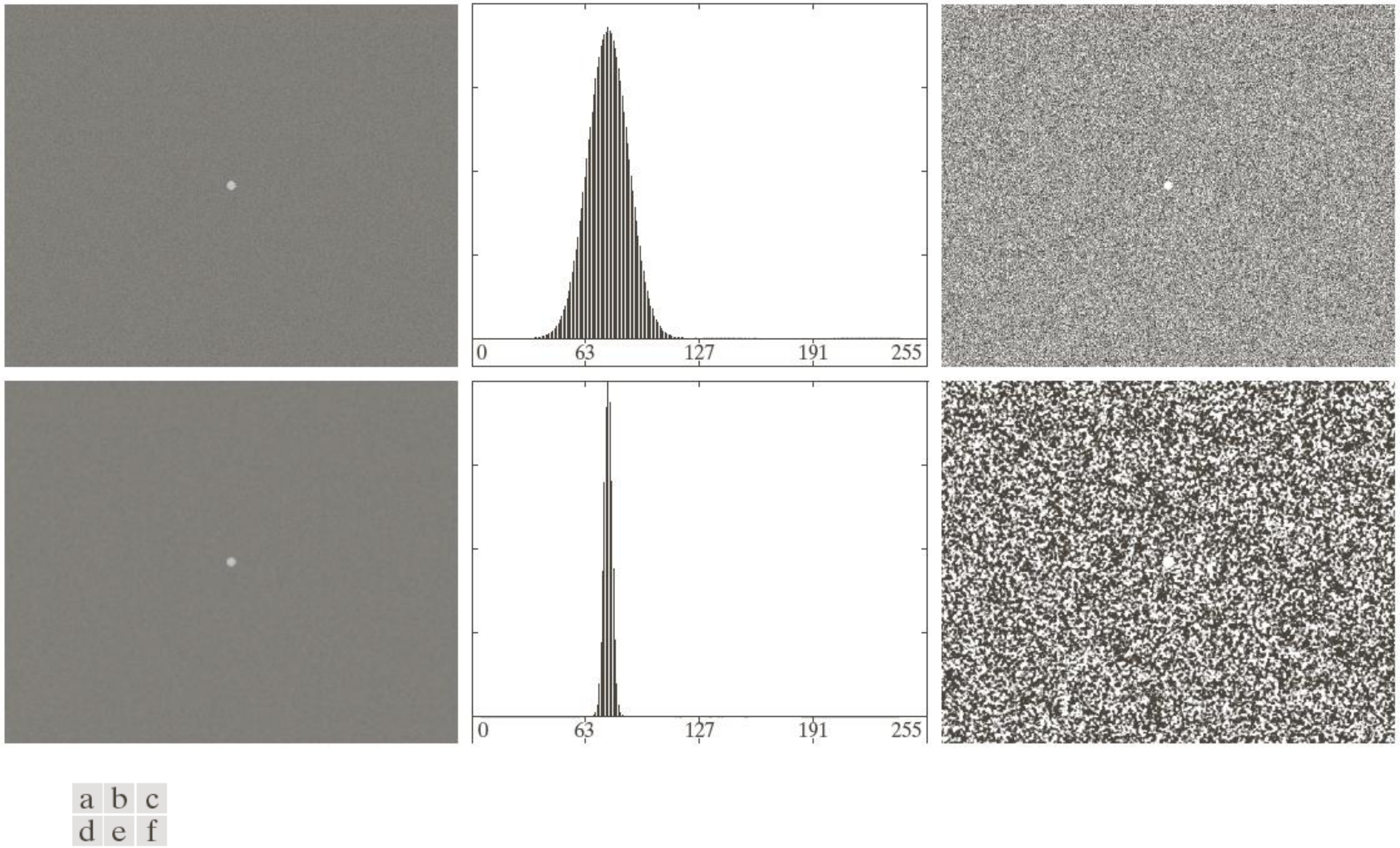


FIGURE 10.41 (a) Noisy image and (b) its histogram. (c) Result obtained using Otsu's method. (d) Noisy image smoothed using a 5 × 5 averaging mask and (e) its histogram. (f) Result of thresholding using Otsu's method. Thresholding failed in both cases.

USING EDGES TO IMPROVE GLOBAL THRESHOLDING

1. Compute an edge image as either the magnitude of the gradient, or absolute value of the Laplacian of f(x,y)
2. Specify a threshold value T
3. Threshold the image and produce a binary image, which is used as a mask image; and select pixels from f(x,y) corresponding to “strong” edge pixels
4. Compute a histogram using only the chosen pixels in f(x,y)
5. Use the histogram from step 4 to segment f(x,y) globally

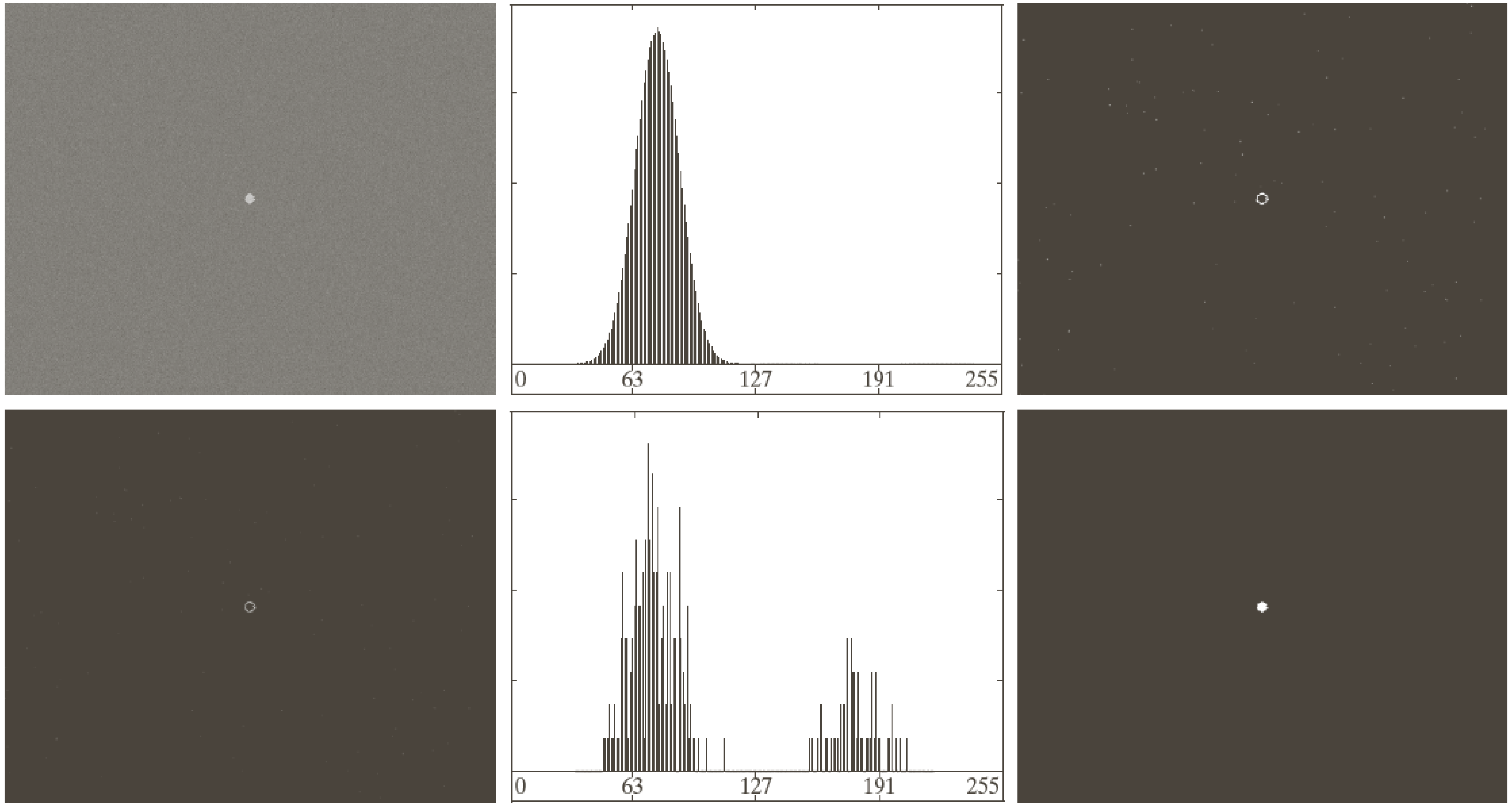
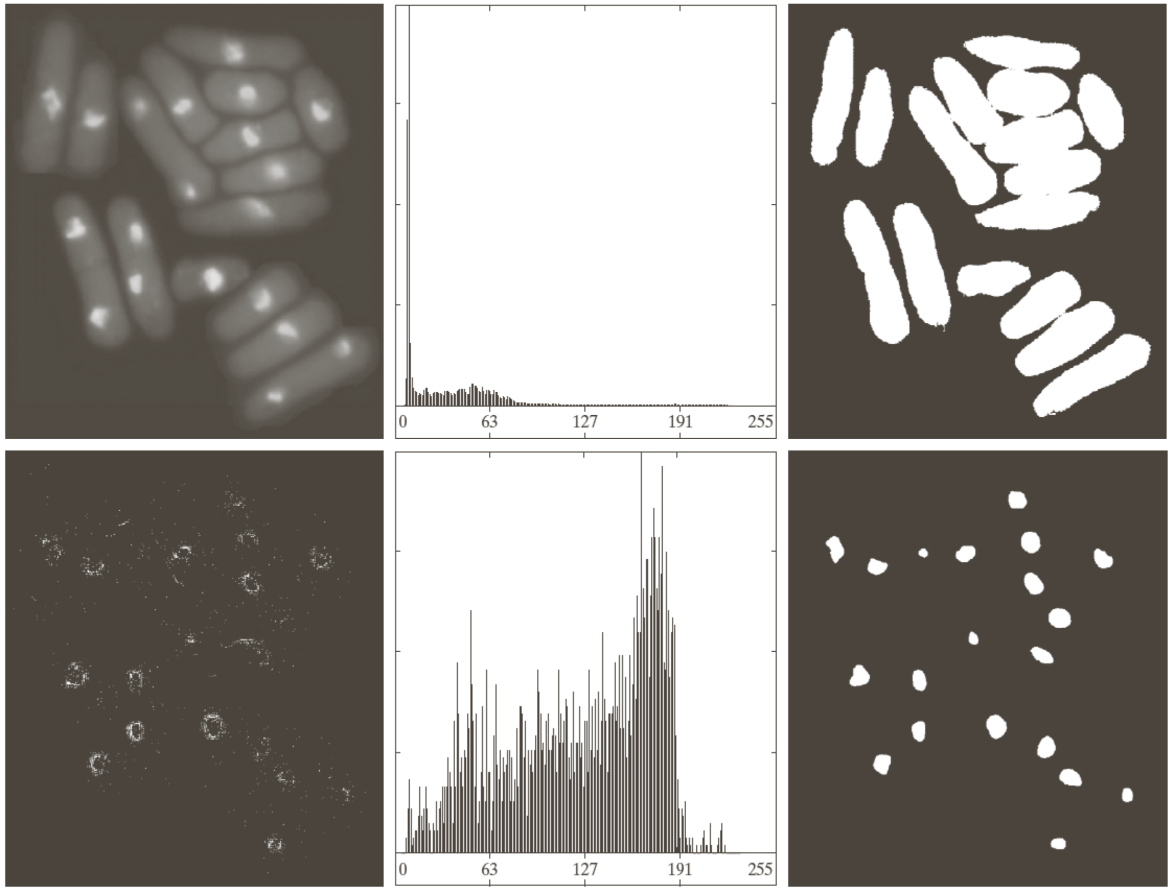


FIGURE 10.42 (a) Noisy image from Fig. 10.41(a) and (b) its histogram. (c) Gradient magnitude image thresholded at the 99.7 percentile. (d) Image formed as the product of (a) and (c). (e) Histogram of the nonzero pixels in the image in (d). (f) Result of segmenting image (a) with the Otsu threshold based on the histogram in (e). The threshold was 134, which is approximately midway between the peaks in this histogram.



a b c
d e f

FIGURE 10.43 (a) Image of yeast cells. (b) Histogram of (a). (c) Segmentation of (a) with Otsu's method using the histogram in (b). (d) Thresholded absolute Laplacian. (e) Histogram of the nonzero pixels in the product of (a) and (d). (f) Original image thresholded using Otsu's method based on the histogram in (e). (Original image courtesy of Professor Susan L. Forsburg, University of Southern California.)

MULTIPLE THRESHOLDS

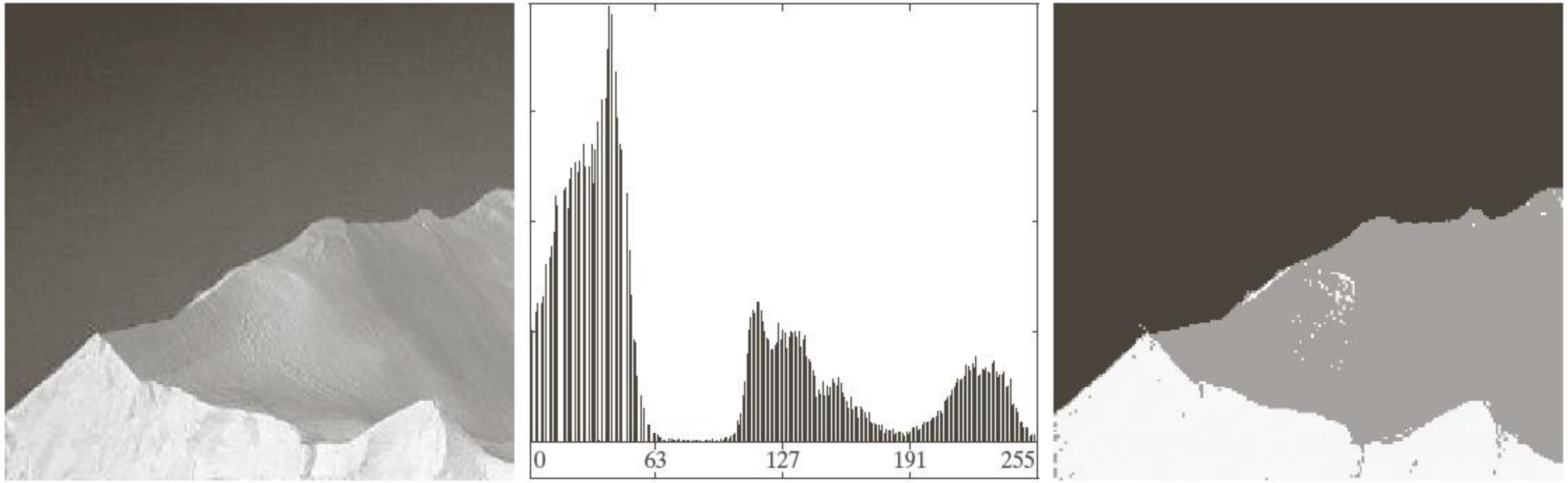
In the case of K classes, $C_1,\ C_2,\ ...,\ C_K$, the between-class variance is

$$\sigma_B^2 = \sum_{k=1}^{K} P_k \left(m_k - m_G\right)^2$$

where $P_k = \sum_{i \in C_k} p_i$ and $m_k = \dfrac{1}{P_k} \sum_{i \in C_k} i p_i$

The optimum threshold values, $k_1^*,\ k_2^*,\ ...,\ k_{K-1}^*$ that maximize

$$\sigma_B^2(k_1^*,\ k_2^*,\ ...,\ k_{K-1}^*) = \max_{0 \le k \le L-1} \sigma_B^2(k_1,\ k_2,\ ...,\ k_{K-1})$$



a b c

FIGURE 10.45 (a) Image of iceberg. (b) Histogram. (c) Image segmented into three regions using dual Otsu thresholds. (Original image courtesy of NOAA.)

VARIABLE THRESHOLDING: IMAGE PARTITIONING

Subdivide an image into nonoverlapping rectangles

The rectangles are chosen small enough so that the illumination of each is approximately uniform.

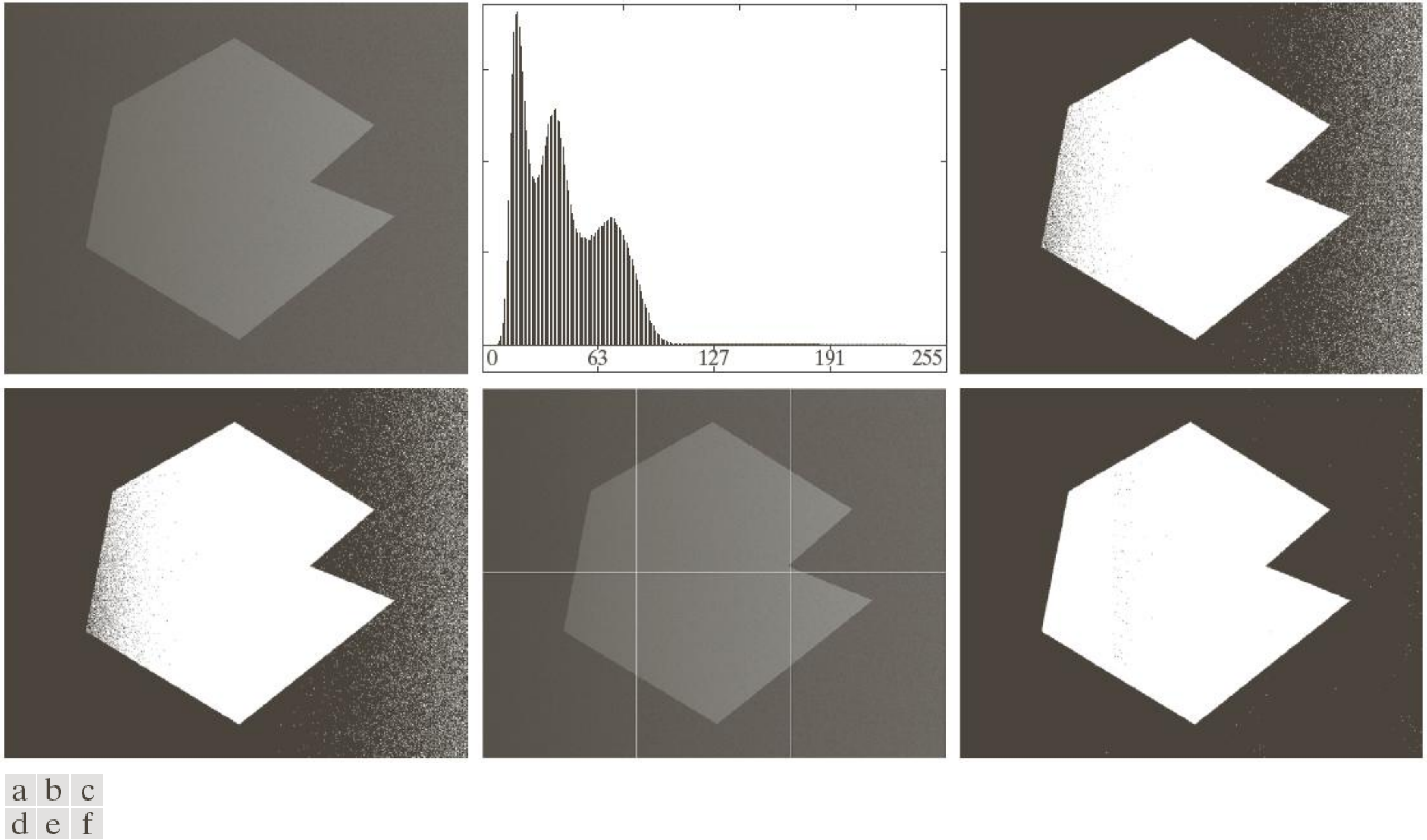


FIGURE 10.46 (a) Noisy, shaded image and (b) its histogram. (c) Segmentation of (a) using the iterative global algorithm from Section 10.3.2. (d) Result obtained using Otsu's method. (e) Image subdivided into six subimages. (f) Result of applying Otsu's method to each subimage individually.

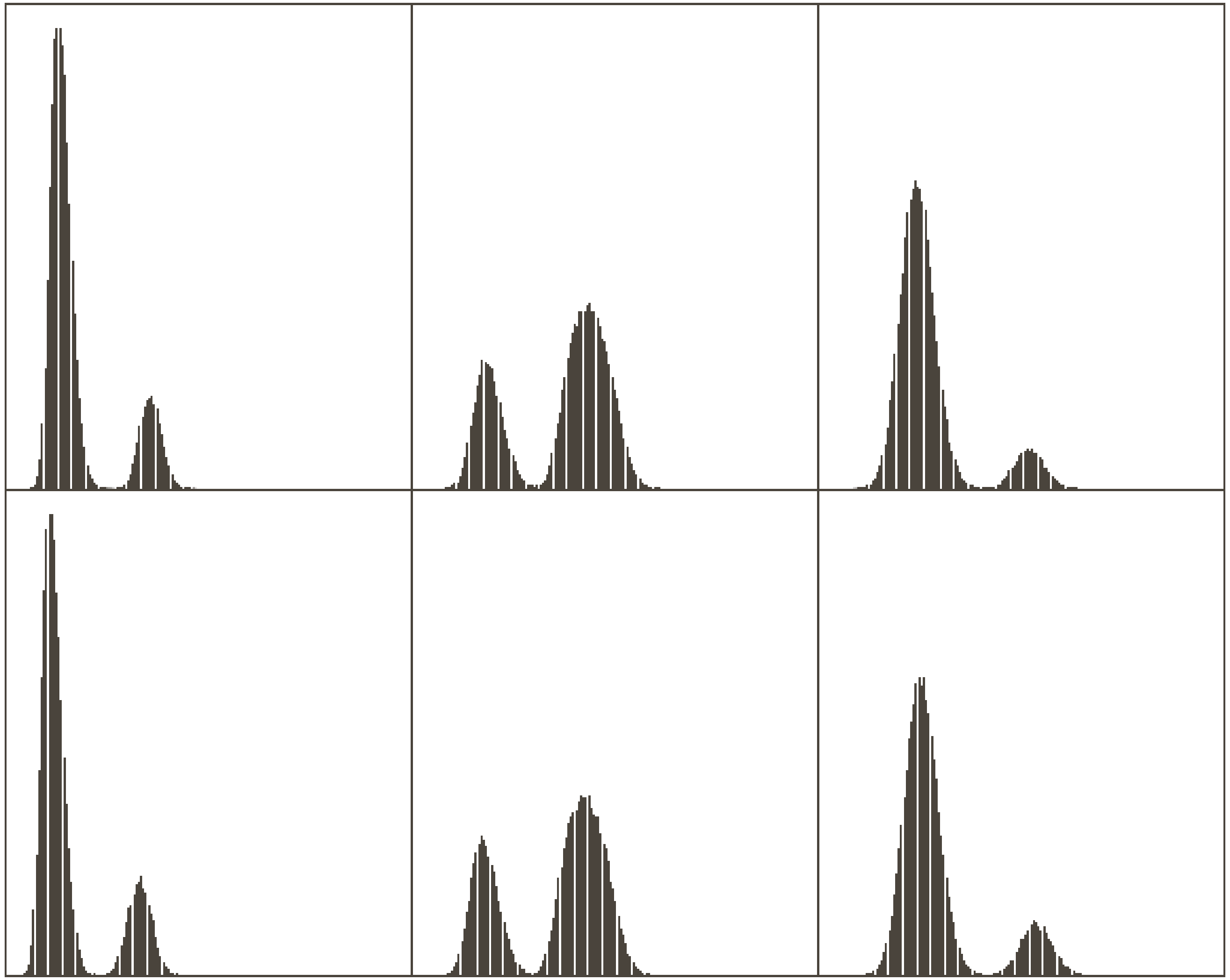


FIGURE 10.47 Histograms of the six subimages in Fig. 10.46(e).

VARIABLE THRESHOLDING BASED ON LOCAL IMAGE PROPERTIES

Let σ_{xy} and m_{xy} denote the standard deviation and mean value of the set of pixels contained in a neighborhood S_{xy}, centered at coordinates (x, y) in an image. The local thresholds,

$$T_{xy} = a\sigma_{xy} + bm_{xy}$$

If the background is nearly constant,

$$T_{xy} = a\sigma_{xy} + bm$$

$$g(x,y) = \begin{cases} 1 & \text{if } f(x,y) > T_{xy} \\ 0 & \text{if } f(x,y) \le T_{xy} \end{cases}$$

A modified thresholding

$$g(x,y) = \begin{cases} 1 & \text{if } Q(\text{local parameters}) \text{ is true} \\ 0 & \text{otherwise} \end{cases}$$

e.g.,

$$Q(\sigma_{xy}, m_{xy}) = \begin{cases} \text{true} & \text{if } f(x,y) > a\sigma_{xy} \text{ AND } f(x,y) > bm_{xy} \\ \text{false} & \text{otherwise} \end{cases}$$

REGION-BASED SEGMENTATION

Region Growing

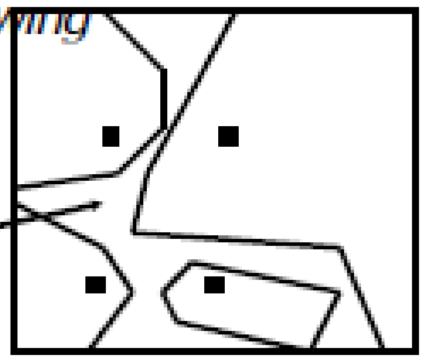
1. Region growing is a procedure that groups pixels or subregions into larger regions.

2. The simplest of these approaches is ***pixel aggregation***, which starts with a set of "seed" points and from these grows regions by appending to each seed points those **neighboring pixels** that have **similar properties** (such as gray level, texture, color, shape).

3. Region growing based techniques are better than the edge-based techniques in noisy images where edges are difficult to detect.

PENDEKATAN REGION-BASED

- *Kekurangannya: belum tentu menghasilkan wilayah-wilayah yang bersambungan*
- *Prosedur:*
  - *Memerlukan criteria of uniformity (kriteria)*
  - *Memerlukan penyebaran seeds atau dapat juga dengan pendekatan scan line*
  - *Dilakukan proses region growing*



unidentified region

REGION GROWING

- Tentukan beberapa piksel seed.
- Seed bisa ditentukan manual atau secara random
- Untuk setiap piksel seed, lihat 4 atau 8 tetangganya, jika kriterianya sama (kriteria bisa berupa perbedaan keabuan dengan seed, dll) maka tetangga tersebut bisa dianggap berada dalam 1 region/daerah dengan piksel seed.
- Teruskan proses dengan mengecek tetangga dari tetangga yang sudah kita cek, dst.
- Tidak bisa hanya digunakan kriteria saja, tanpa melihat konektivitas ketetanggaan, karena bisa tidak membentuk daerah
- Stopping rule kadang tidak mencakup semua kemungkinan sehingga pada akhir region growing ada piksel yang belum dicek sama sekali.

Suppose that we have the image given below.

(a) Use the region growing idea to segment the object. The seed for the object is the center of the image. Region is grown in horizontal and vertical directions, and when the difference between two pixel values is less than or equal to 5.

Table 1: Show the result of Part (a) on this figure.

10	10	10	10	10	10	10
10	10	10	69	70	10	10
59	10	60	64	59	56	60
10	59	10	<u>60</u>	70	10	62
10	60	59	65	67	10	65
10	10	10	10	10	10	10
10	10	10	10	10	10	10

Suppose that we have the image given below.

(a) Use the region growing idea to segment the object. The seed for the object is the center of the image. Region is grown in horizontal and vertical directions, and when the difference between two pixel values is less than or equal to 5.

Table 1: Show the result of Part (a) on this figure.

10	10	10	10	10	10	10
10	10	10	69	70	10	10
59	10	60	64	59	56	60
10	59	10	<u>60</u>	70	10	62
10	60	59	65	67	10	65
10	10	10	10	10	10	10
10	10	10	10	10	10	10

4-connectivity

Suppose that we have the image given below.

(a) Use the region growing idea to segment the object. The seed for the object is the center of the image. Region is grown in horizontal and vertical directions, and when the difference between two pixel values is less than or equal to 5.

Table 1: Show the result of Part (a) on this figure.

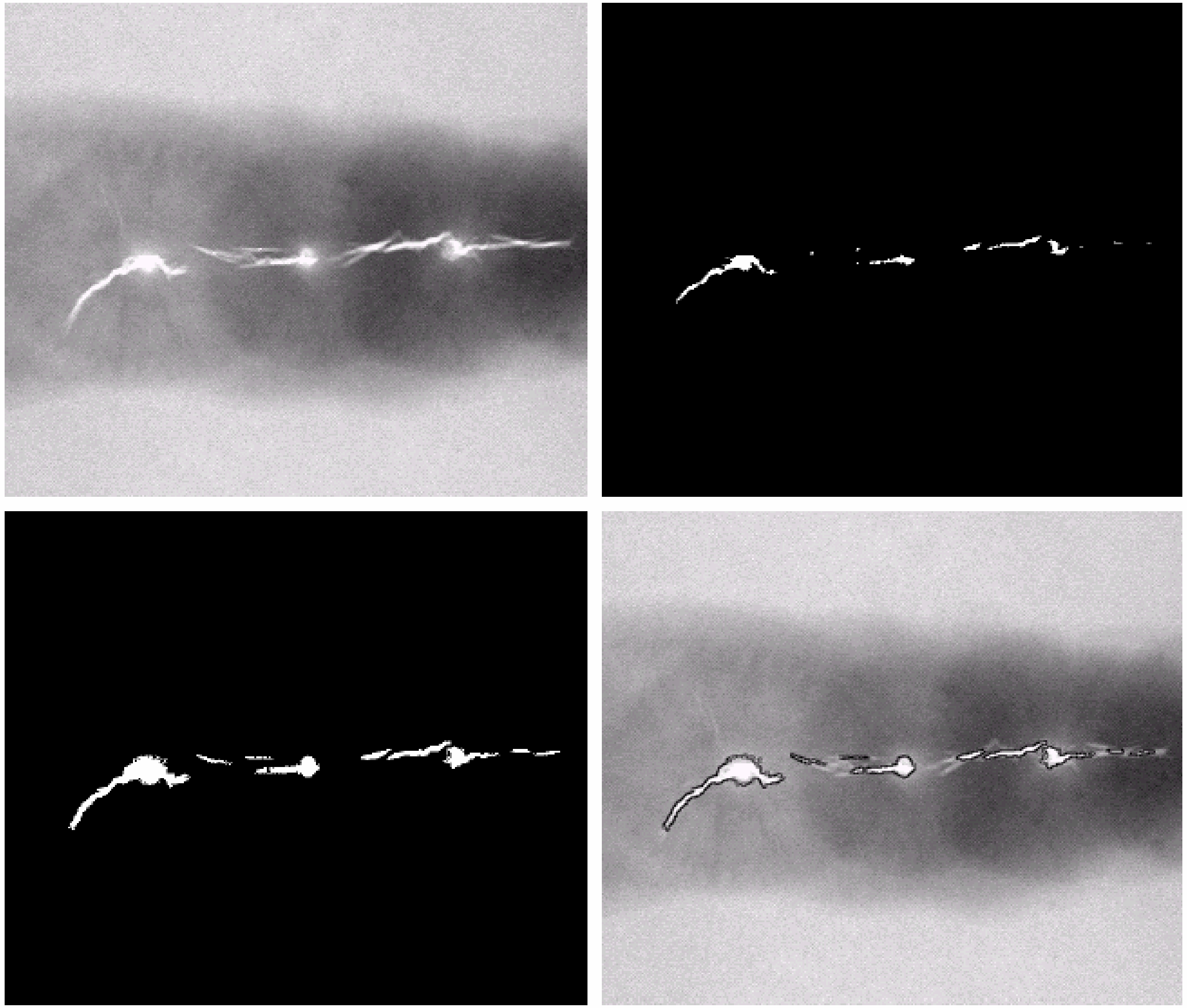
10	10	10	10	10	10	10
10	10	10	69	70	10	10
59	10	60	64	59	56	60
10	59	10	60	70	10	62
10	60	59	65	67	10	65
10	10	10	10	10	10	10
10	10	10	10	10	10	10

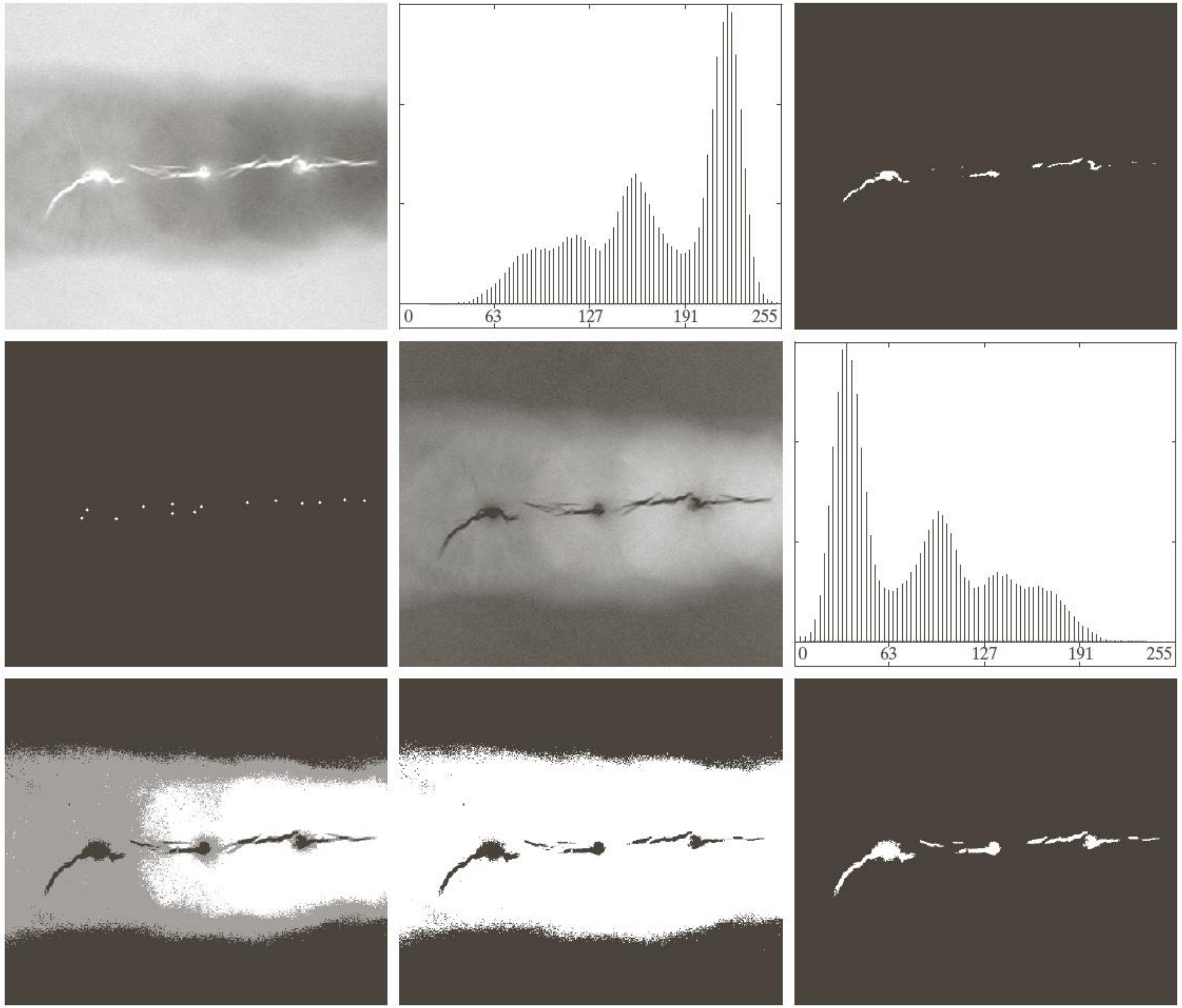
8-connectivity

CONTOH

a b
c d

FIGURE 10.40
(a) Image showing defective welds. (b) Seed points. (c) Result of region growing. (d) Boundaries of segmented defective welds (in black). (Original image courtesy of X-TEK Systems, Ltd.).





a b c
d e f
g h i

FIGURE 10.51 (a) X-ray image of a defective weld. (b) Histogram. (c) Initial seed image. (d) Final seed image (the points were enlarged for clarity). (e) Absolute value of the difference between (a) and (c). (f) Histogram of (e). (g) Difference image thresholded using dual thresholds. (h) Difference image thresholded with the smallest of the dual thresholds. (i) Segmentation result obtained by region growing. (Original image courtesy of X-TEK Systems, Ltd.)

REGION SPLITTING N MERGING

Splitting: membagi citra menjadi beberapa daerah berdasarkan kriteria tertentu (teknik quadtree)

Merging: gabungkan daerah-daerah berdekatan yang memiliki kriteria yang sama.

Kriteria: bisa varian keabuan dll

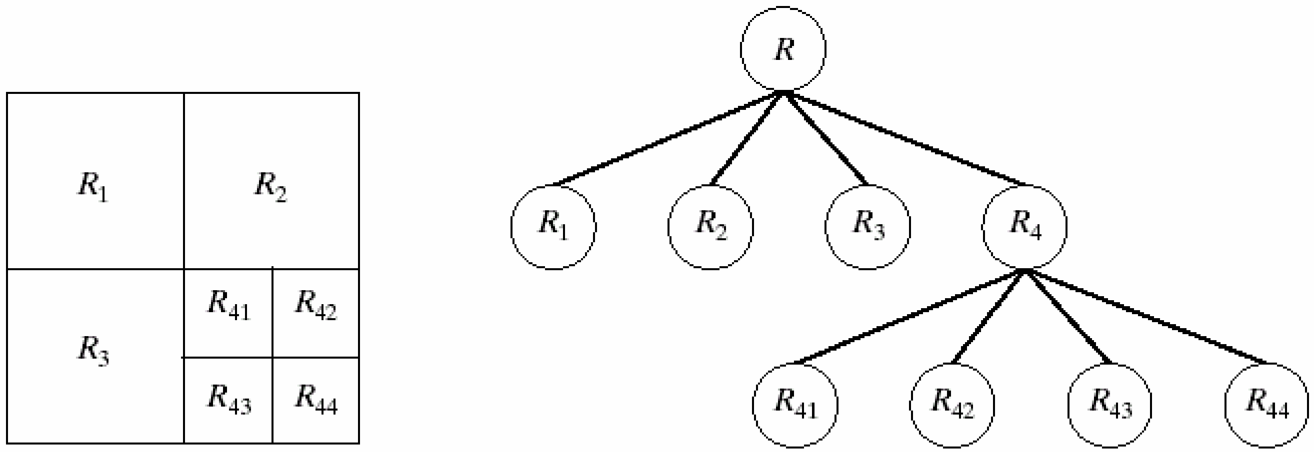
Prosedur umum:

- Split R menjadi 4 kuadran disjoint jika P(R) = FALSE
- Merge sembarang daerah berdekatan Ri, Rj jika $P(R_i \cup R_j)$ =TRUE
- Berhenti jika tidak ada proses split n merge yang bisa dilakukan

CONTOH

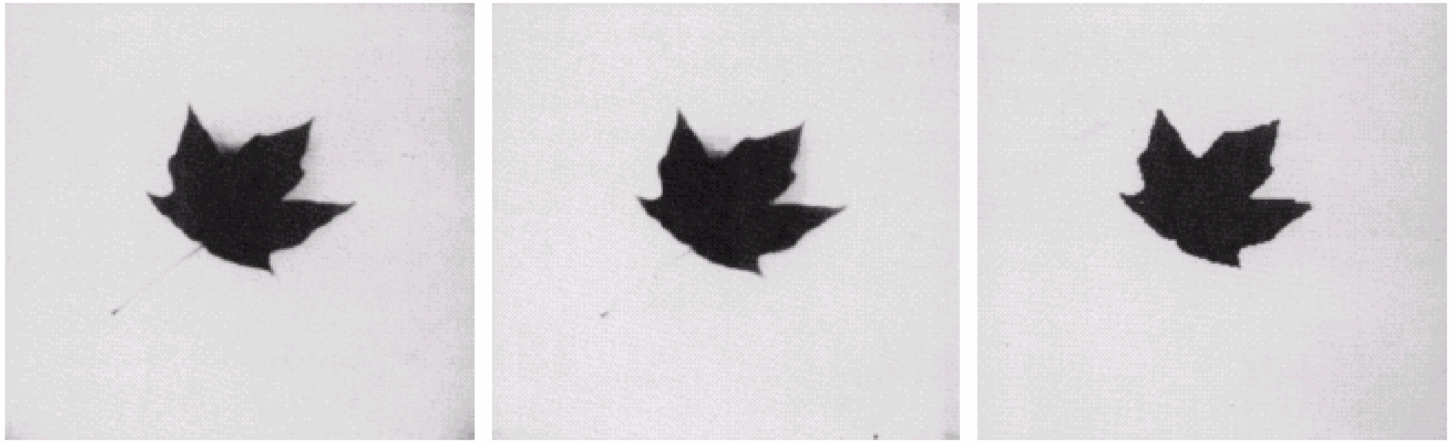
a b

FIGURE 10.42
(a) Partitioned image.
(b) Corresponding quadtree.



a b c

FIGURE 10.43
(a) Original image. (b) Result of split and merge procedure.
(c) Result of thresholding (a).



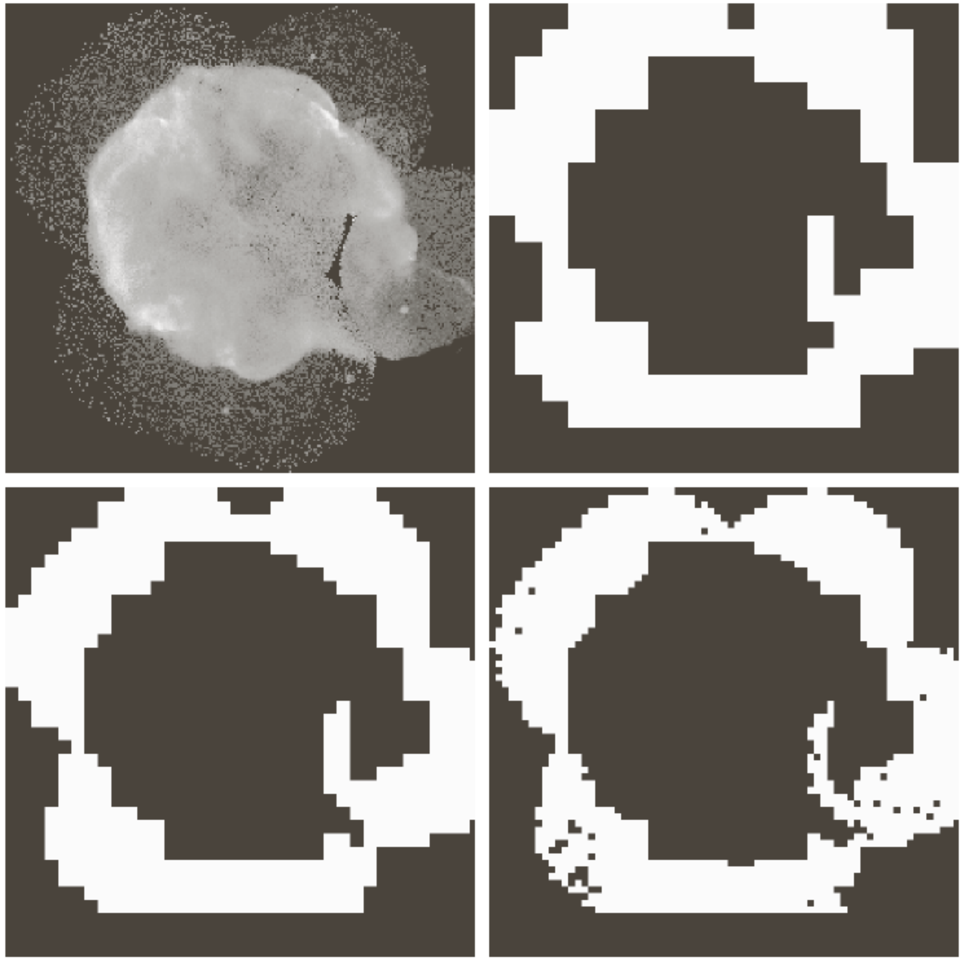
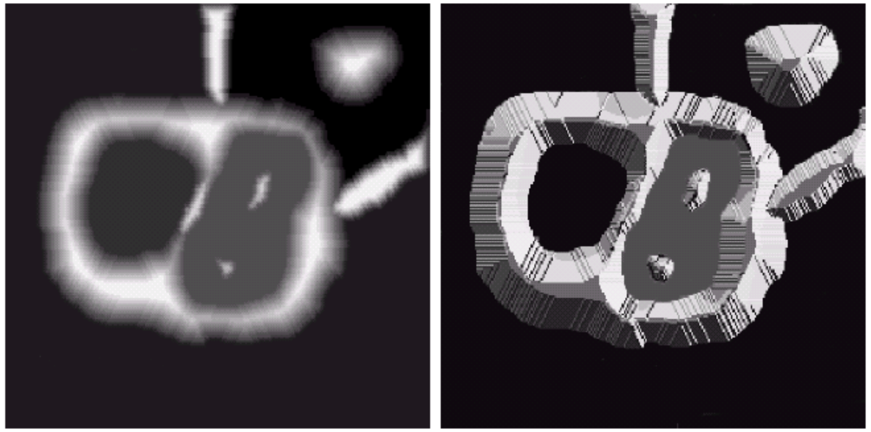


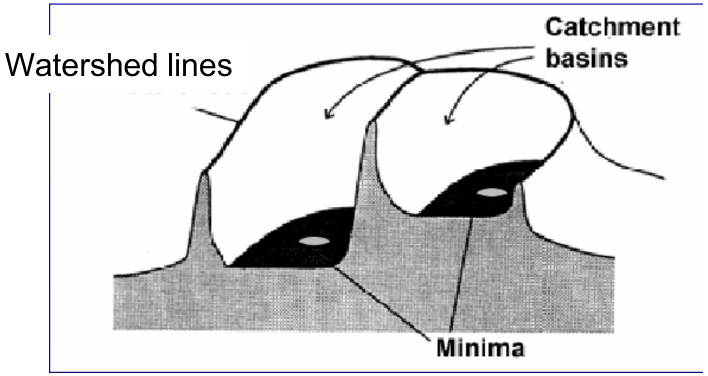
FIGURE 10.53 (a) Image of the Cygnus Loop supernova, taken in the X-ray band by NASA's Hubble Telescope. (b)–(d) Results of limiting the smallest allowed quadregion to sizes of 32×32, 16×16, and 8×8 pixels, respectively. (Original image courtesy of NASA.)

SEGMENTASI MENGGUNAKAN MORPHOLOGICAL WATERSHEDS

Tiga jenis titik pada intepretasi topographic:

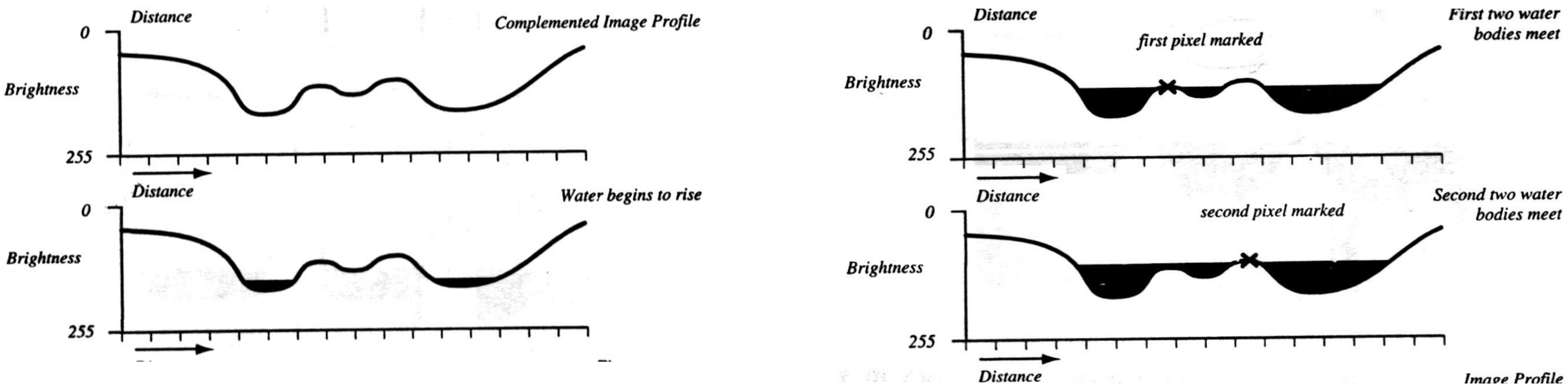
- Poin milik minimum regional
- Titik di mana setetes air akan jatuh ke minimum tunggal. (→Daerah tangkapan atau watershed)
- Titik di mana setetes air akan sama-sama mungkin jatuh ke lebih dari satu minimum. (→Garis pemisah atau *watershed lines*.)

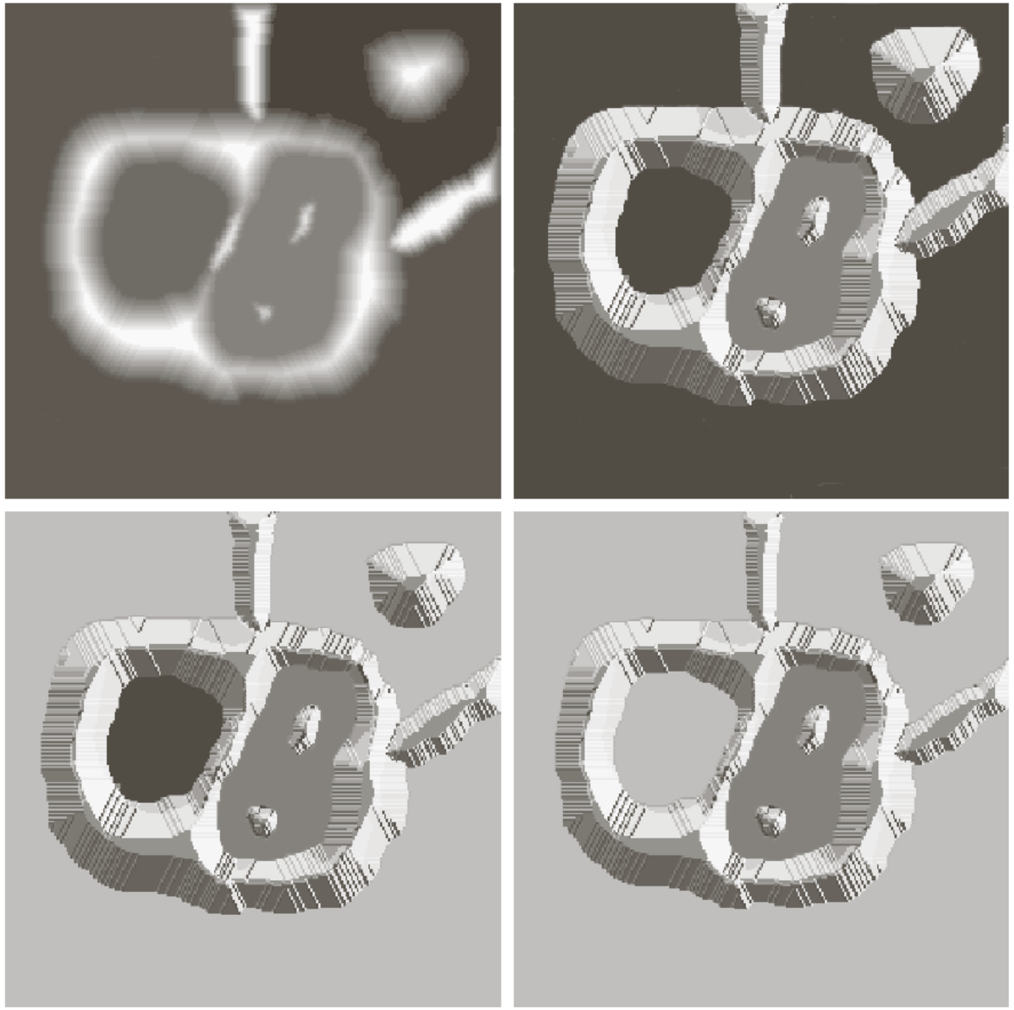




WATERSHED SEGMENTATION: EXAMPLE

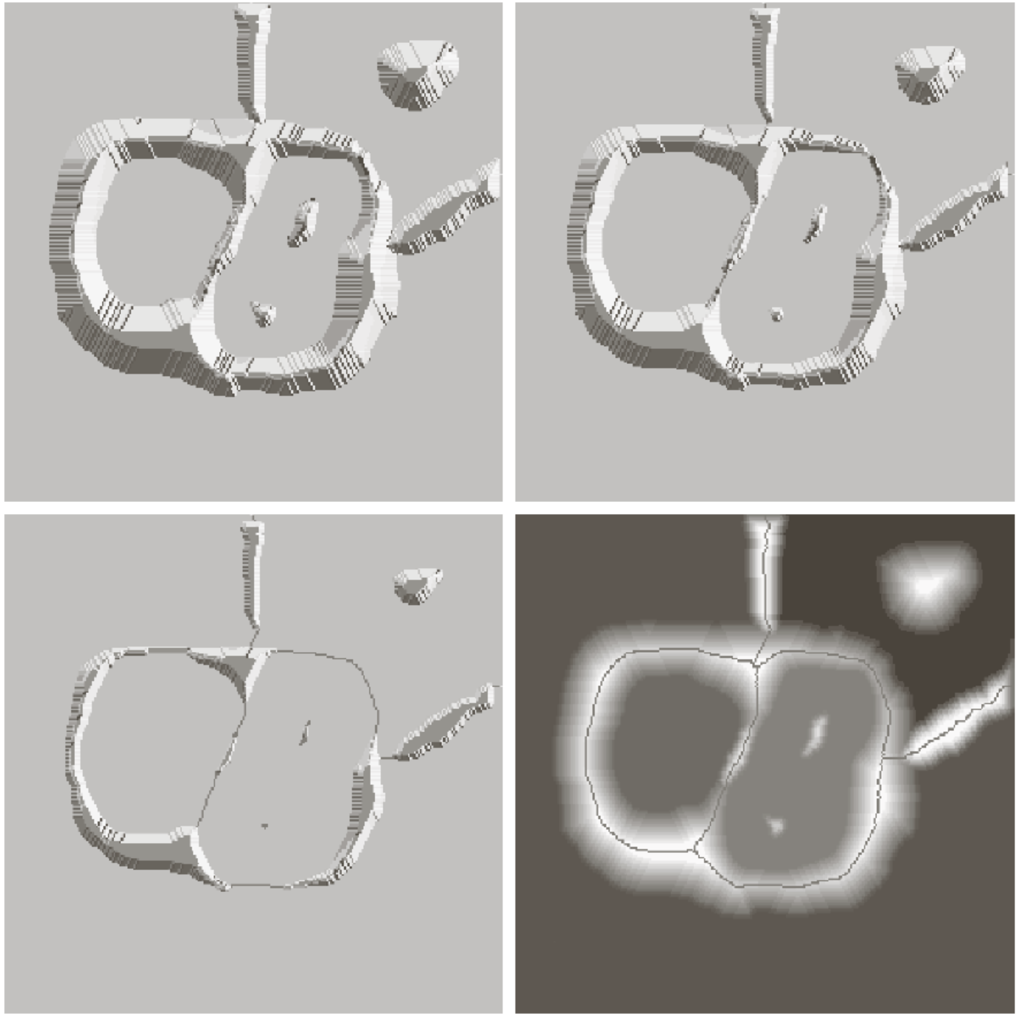
- Tujuan adalah melakukan pencarian garis watershed
- The idea is simple:
  - Misalkan sebuah lubang dilubangi di setiap minimum regional dan seluruh topografi dibanjiri dari bawah dengan membiarkan air naik melalui lubang dengan laju yang seragam.
  - Ketika air naik di DAS yang berbeda adalah tentang penggabungan, bendungan dibangun untuk mencegah penggabungan. Batas bendungan ini sesuai dengan garis DAS.





a	b
c	d

FIGURE 10.54
(a) Original image.
(b) Topographic view. (c)–(d) Two stages of flooding.



e	f
g	h

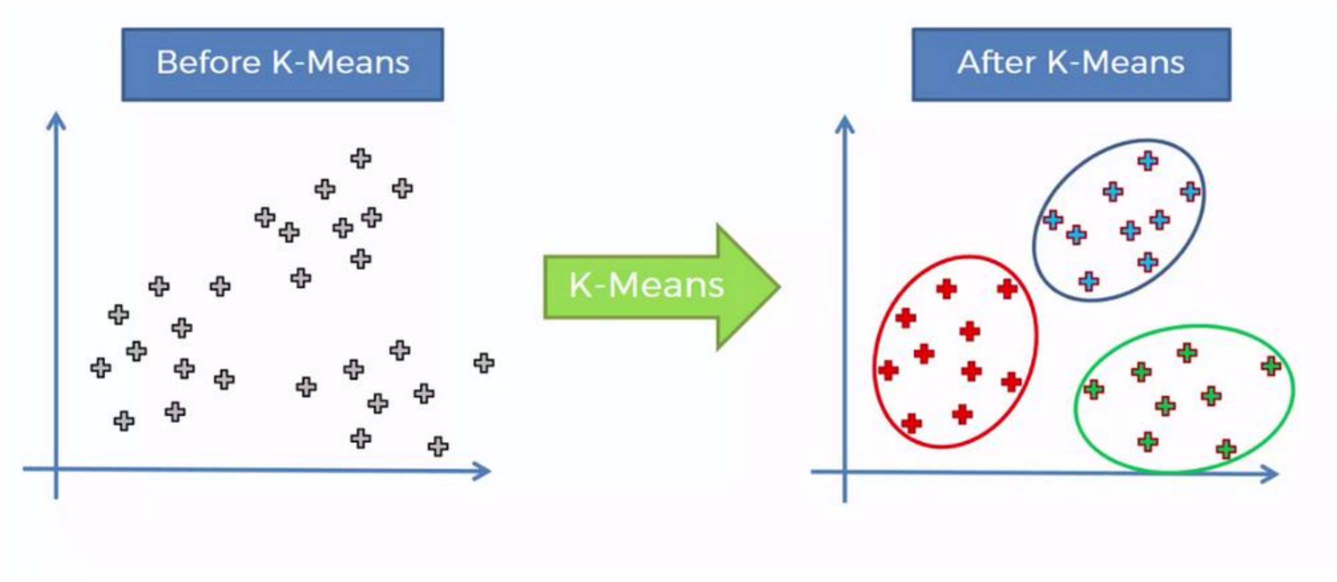
FIGURE 10.54 *(Continued)* (e) Result of further flooding. (f) Beginning of merging of water from two catchment basins (a short dam was built between them). (g) Longer dams. (h) Final watershed (segmentation) lines. (Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)

ALGORITMA SEGMENTASI WATERSHED

- Mulailah dengan semua piksel dengan nilai serendah mungkin.
  - Ini membentuk dasar untuk DAS awal Start with all pixels with the lowest possible value.
- Untuk setiap tingkat intensitas k:
  - Untuk setiap kelompok piksel dengan intensitas k
    1. Jika berdekatan dengan tepat satu wilayah yang ada, tambahkan piksel ini ke wilayah itu
    2. Jika berdekatan dengan lebih dari satu wilayah yang ada, tandai sebagai batas
    3. Jika tidak, mulailah wilayah baru

SEGMENTASI DENGAN K-MEANS CLUSTERING

- *K-Means Clustering* merupakan metode *unsupervised clustering* yang mengelompokkan titik data ke *k* kluster berdasarkan jarak dengan pusat kluster. Titik data dikelompokkan sekitar pusat kluster (*centroid*)



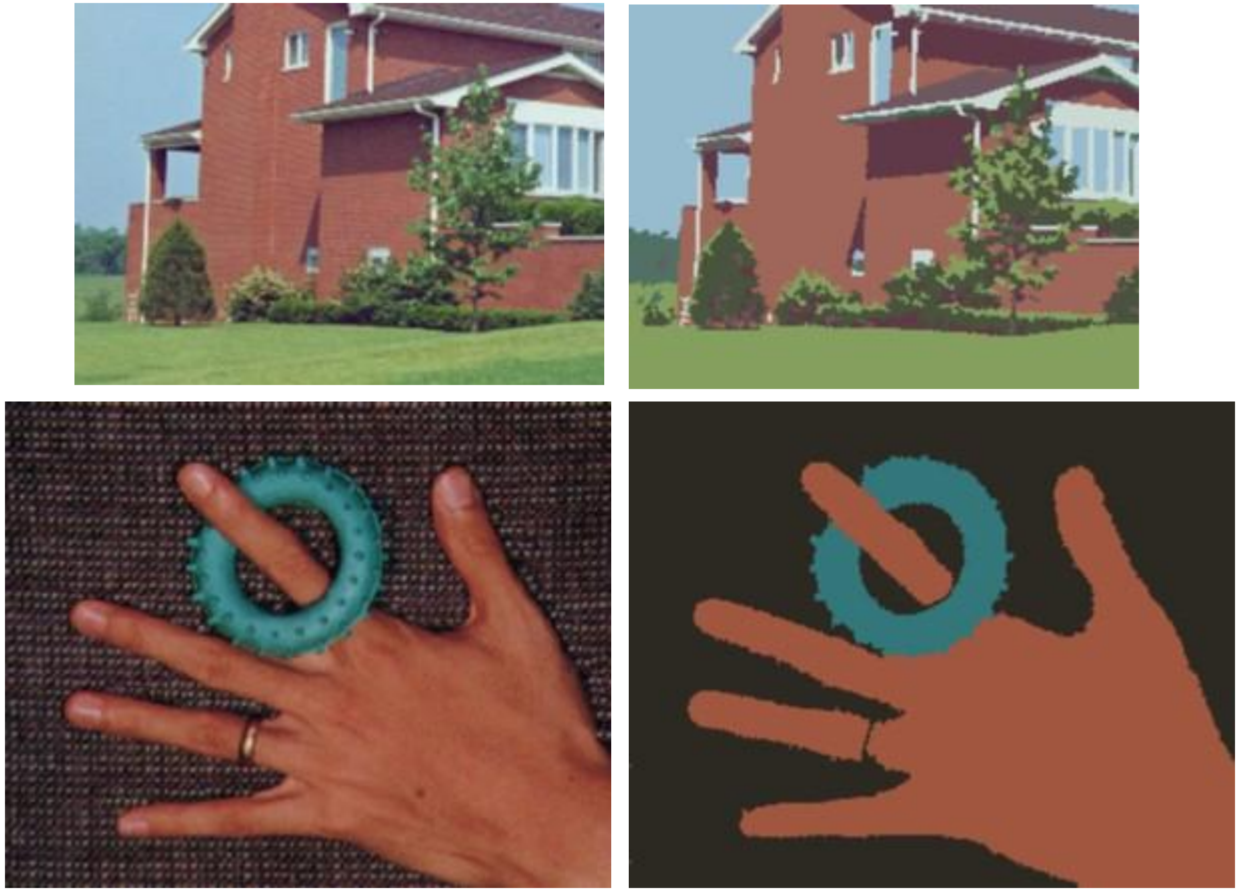
https://medium.com/data-science-and-learning/k-means-clustering-4a700d4a4720

SEGMENTASI MENGGUNAKAN *K-MEANS*

1. Menentukan jumlah *cluster* pada citra hasil *Pre-processing*
2. Menghitung nilai *centroid* secara acak.
3. Menghitung jarak *pixel* ke *centroid* dan mengelompokkan *pixel* berdasarkan jarak terdekat.
4. Menghitung rata-rata dari masing-masing data cluster untuk menghasilkan centroid baru.
5. Mengelompokkan kembali *pixel* sesuai dengan *centroid baru* tersebut.
6. Apabila masih terdapat *pixel* yang berpindah *cluster* maka dilakukan perhitungan *centroid* baru, tetapi apabila jika sudah tidak ada nilai *pixel* yang berpindah *cluster* maka proses *clustering* berakhir

SEGMENTASI DENGAN K-MEANS

- Contoh



D. Comaniciu and P. Meer, *Robust Analysis of Feature Spaces: Color Image Segmentation*, 1997.


Thank you!!